AI Strategic Autonomy Framework

Table of Contents

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Document Package — Framework, Field Guides, State of Play, and Evidence Base

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At a glance

What is this?

The AI Strategic Autonomy Framework (AI-SAF) is a way to measure AI sovereignty — whether a government or a company can keep its important AI systems running, and keep improving them, if an outside supplier or technology cuts off access. It is not a ranking of who is "best at AI." It measures something different: how much keeps working, and keeps developing, when access changes.

Two questions, two scores

Every assessment answers "What still works if a supplier cuts us off tomorrow?" (resilience) and "Who gets to decide our technology path?" (autonomy). From v0.7.1, these are no longer just an interpretive lens: every sub-indicator is tagged R (resilience) or A (autonomy), and each assessment reports the two subtotals alongside the headline score, which runs from 0 to 100 (higher = more sovereign). A subject can score high on one and low on the other; true sovereignty needs both.

Two versions, shared building blocks

Governments and companies face the same pressures but react differently, so the framework keeps two versions instead of one combined score. Both share eight pillars: Data, Models, Infrastructure, Hardware, Manufacturing, Software, Human Capital, and Regulatory. **AI-SAF-C (business)** — the lead variant of this package — is for companies; it weights what companies control (cloud choice, model choice, compliance, speed) and adds a ninth pillar, Culture and Operating Model. **AI-SAF-N (national)** is for governments and blocs; it weights what governments control — chips, sovereign clouds, chip factories, and rule-making. The same pillar can mean opposite things at the two levels — the full argument is in Framework §1.2, and it is the reason a single combined score cannot work.

How does scoring work?

Pillars break into sub-indicators, scored on two angles: capability (what is actually in place today — the basis for the main score) and commitment (what has only been promised — used to estimate the position three years out). Modifiers then adjust the scores: AI-SAF-N has four (rule-stability, geopolitical-alliance, critical-minerals, operating-model), applied in a fixed order set out in the National Field Guide; AI-SAF-C applies a scale discount for smaller firms. Each dependency is labelled a bottleneck (replaceable) or a structural constraint (no realistic replacement). Scores from different levels (national vs. corporate) are not comparable.

The central idea — Version Deadlock

The framework's most original concept is the trap of being unable to move to the next version of a technology you depend on. Needing a supplier today is manageable; losing the ability to keep up is slow, hard to spot, and often impossible to undo. Outcomes fall into four states — Recoverable, Slow-clock, Vulnerable-but-recoverable, and Terminal — with one caution added in this revision: a deadlock ended by the supplier's side (restoration) says nothing about the subject's own position. The 2026 Fable 5 / Mythos 5 suspension and 19-day restoration is now the framework's flagship live case.

Putting it to use — gap analysis

The framework's main job is not the score but pointing to where the next investment does the most good: score today, project three years out if nothing changes, work out the payoff per unit of investment for each pillar, then choose the best mix. For companies, people come first, then reaching L2 on Models (adapting an open model). For countries, infrastructure and people pay off most, over 5—10 years.

Package contents

- **Document 1 · The Framework** — theory, methodology, terminology, and the merged glossary.
- **Document 2 · AI-SAF-C Field Guide** (business) — nine pillars with scoring rubrics, the BNP Paribas worked example with an ING calibration check, and the crosswalk to NIST AI RMF, ISO/IEC 42001, and GAIA-X.
- **Document 3 · AI-SAF-N Field Guide** (national) — eight pillars with scoring rubrics, the four modifiers, worked examples for Bulgaria, France, and the UAE, and the INSAIT/BgGPT case study.
- **Document 4 · State of Play** — the dated market snapshot (models, clouds, chips, manufacturing, software), separated from the rubrics so it can be refreshed each revision without touching the scoring.
- **Document 5 · Evidence Base** — sovereignty incidents and sources, 2024—2026, updated to August 2026.

What is uncertain

Part V of the Framework lists the open questions candidly, updated for this revision: how the EU AI Act's split timeline plays out (GPAI enforcement live since 2 August 2026; high-risk deferred); where the Cloud and AI Development Act's sovereignty framework lands in trilogue; whether hyperscalers' own chips loosen NVIDIA's grip; whether export controls on model access (the Fable 5 / Mythos 5 episode) become a repeated instrument; life after the Intel Magdeburg cancellation; how much announced funding is actually spent; and the limits of the framework's own additive scoring.

About this package

The AI Strategic Autonomy Framework (AI-SAF) assesses AI sovereignty — the capacity of a state or a firm to keep its critical AI systems running, and to keep evolving them, when external access to suppliers or technology is constrained. The framework decomposes sovereignty along two axes (resilience and autonomy), across eight shared pillars, with separate weightings for states and firms, and with an explicit account of the Version Deadlock condition — being unable to evolve because access to the next version of a key technology has been interrupted.

The two field guides are self-contained. Each opens with a short bridge to the Framework. Reading the Framework first is recommended but not required. Dated market material lives in Document 4 (State of Play) and is refreshed each revision; the rubrics in the field guides are designed to stay stable between revisions.

A note on the acronym. AI-SAF here stands for AI Strategic Autonomy Framework. It is not the UK AI Security Institute or the US CAISI, which are referenced inside the documents but are separate institutions. If this framework moves toward external publication, rebranding is an open question.

What kind of feedback would be most useful?

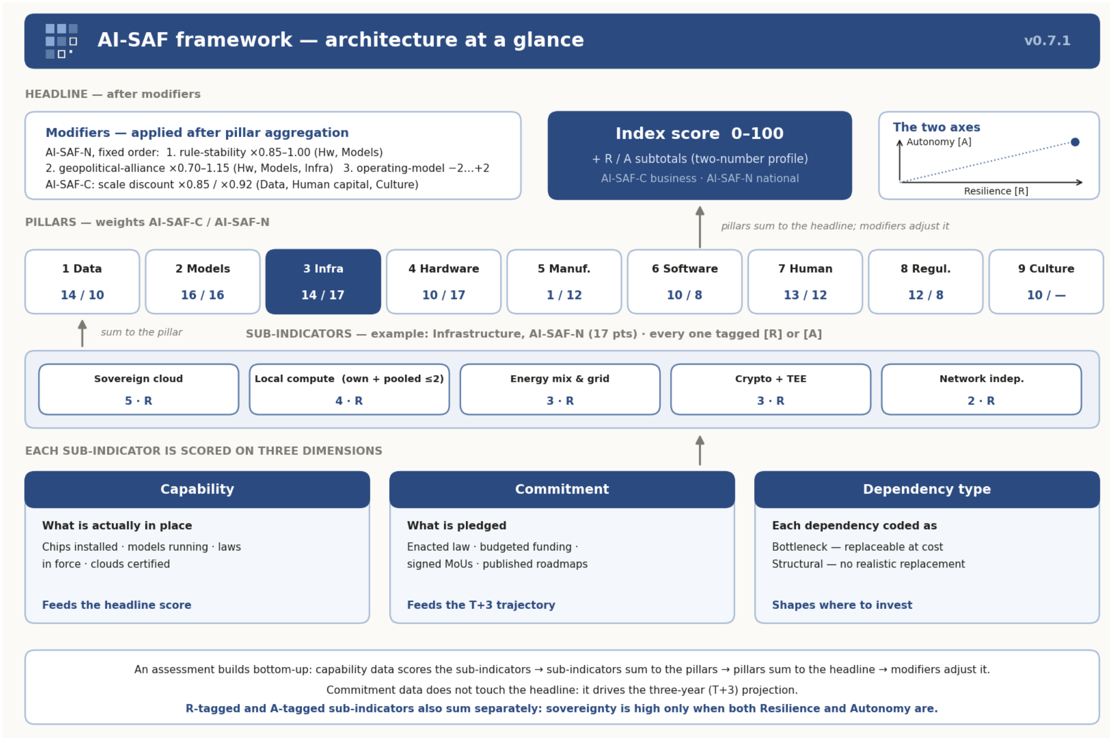
The framework was first circulated in May 2026; the current revision is v0.7.1 (August 2026). Feedback is most valuable on four points: whether the pillar weights are defensible against the 2024–2026 incident record; whether Version Deadlock is a genuinely useful concept or repackaged lock-in; whether the worked-example profiles (Bulgaria ~46, France ~70, UAE ~57 on AI-SAF-N; BNP Paribas ~62 and ING ~47 on AI-SAF-C) track your own read of these subjects; and whether Part V lists the right uncertainties. A half-page of honest skepticism is more useful than polite acknowledgment.

Document 1 · The Framework

Conceptual foundation — theory, methodology, terminology. Version 0.7.1 · August 2026 ·

How to read this document

This framework uses a specific vocabulary and a specific scoring structure throughout. This section introduces both. A reader already familiar with sovereignty indices may skim it; a reader new to the framework should read it carefully.



The AI-SAF framework architecture. An assessment builds from the bottom up: capability and commitment data score the sub-indicators, sub-indicators score the pillars, pillars sum to the headline, modifiers adjust it. Every sub-indicator carries an R (resilience) or A (autonomy) tag; the two subtotals are reported alongside the headline.

The pillars and weights

Every AI-SAF assessment — a company under AI-SAF-C or a country under AI-SAF-N — scores the subject on eight shared pillars. A ninth, Culture and Operating Model, applies to companies only.

Pillar	What it measures	AI-SAF-C	AI-SAF-N
1. Data	Language and proprietary data, where data is kept, the laws covering it, and what leaves at inference time	14	10
2. Models	Access to general-purpose AI models, the ability to adapt them, and a disciplined portfolio across providers and jurisdictions	16	16
3. Infrastructure	Cloud you control, data-centre capacity, reliable power, and confidential computing	14	17
4. Hardware	Access to AI chips, more than one chip design, and reserves	10	17
5. Manufacturing	Chip-factory capacity, advanced packaging,	1	12

	critical minerals, and the supply chain behind them		
6. Software	The AI software stack, CUDA dependence, and how much of the tool set you control	10	8
7. Human Capital	Researchers, engineers, training pipelines, and retention	13	12
8. Regulatory	Following the rules thoroughly (companies) or the power to make them (countries)	12	8
9. Culture and Operating Model	How decisions get made and how disciplined operations are (companies only)	10	—
Total		**100**	**100**

The two axes — and how they reach the score

Each pillar score feeds two separate qualities. **Resilience**: can you keep your important AI systems running if an outside supplier cuts off access? **Autonomy**: do you get to decide your own technology path? The two are separate — a subject can hold a full working copy of a model (high resilience) yet be unable to improve it without outside help (low autonomy), and the reverse.

From v0.7.1 this distinction is carried into the arithmetic. Every sub-indicator in both field guides is tagged **[R]** or **[A]**. The headline score is still the single 0–100 sum, but each assessment also reports the R-subtotal and A-subtotal as a two-number profile. Full definitions and practical tests are in §1.1.

Scoring

The **main score** runs 0 to 100; higher means more sovereign. It is the eight (or nine) pillar scores added together, before modifiers. Each pillar's maximum equals its weight. **Sub-indicator scores** break each pillar into three to six parts, each worth a set number of points; the Field Guides give every rubric.

Three things sit alongside the pillar scores:

- **Modifiers** are adjustments applied after the pillar scores are worked out. AI-SAF-N has four — rule-stability (×0.85–1.00 on Hardware and Models), geopolitical-alliance (×0.70–1.15 on Hardware, Models, and Infrastructure), operating-model (±2 on the headline), and critical-minerals (scored inside Manufacturing) — applied in the fixed order given in the National Field Guide, Part II. AI-SAF-C applies a scale discount to three pillars for firms under 2,000 employees (×0.85 below 500 employees; ×0.92 from 500 to 2,000).
- **Capability and commitment** are the two angles from which each sub-indicator is scored. Capability is what is actually in place today, judged against hard facts; it feeds the main score. Commitment is what has been promised — budgets, policies, laws — and feeds the three-year projection.

- **Dependency type** labels each dependency as a bottleneck (replaceable at reasonable cost) or a structural constraint (no realistic replacement). It does not change the score; it shows which dependencies are worth spending money to fix.

Levels of analysis

AI-SAF-C is for a single company; AI-SAF-N for a country or bloc (the EU can be scored as a group). A sector score is just the AI-SAF-C scores of the sector's biggest players combined. Product-level scoring is an internal management tool, not a comparison instrument. **Scores from different levels are not comparable**: a national 72 is not "higher" than a corporate 68 — they measure related but different things.

Executive summary

AI sovereignty has many parts, so one scoring system cannot fairly judge both governments and companies. Governments set the rules and prices in chip diplomacy; companies follow those rules and take the prices they are given. Governments plan ten to twenty years out; companies three to five. Governments answer to citizens; companies to shareholders and regulators. Because each side reacts differently to the same pressure, a single score for both looks exact but means little. The framework therefore uses two versions sharing the same eight pillars: **AI-SAF-C** for companies (weighting cloud choice, model choice, compliance depth, culture, and speed, with a ninth Culture pillar) and **AI-SAF-N** for governments (weighting chip access, sovereign cloud, fabs, and rule-making).

The framework measures resilience ("what still works if a supplier cuts us off tomorrow?") and autonomy ("who decides our technology path?"), reported as a two-number profile alongside the headline score. Its most original idea — Version Deadlock, the trap of losing the ability to move to the next version of a technology you depend on — is developed in Part III, where the 2026 Fable 5 / Mythos 5 suspension now serves as the first live AI case, from cut-off through restoration.

Part I. Fundamental concepts

1.1 A two-dimensional definition

AI sovereignty is measured on two separate scales, each scored on its own.

Resilience. Can you keep your important AI systems running if an outside supplier cuts off some or all of your access? The key question: "What still works if a supplier cuts us off tomorrow?" A simple test: if the company behind a top AI model shut off your access tomorrow, how much of your work would keep going? That depends on whether you have a backup model you can run yourself, and whether your systems are wired so tightly to one vendor's way of working that nothing else fits. Think of the power grid: a resilient grid keeps the lights on when one supplier fails, dimming gradually instead of going dark.

Every major AI-sovereignty event of 2024—2026 was a test of this scale — Microsoft France's CLOUD Act testimony (June 2025), the H20 revenue-share deal (August 2025), the DeepSeek bans (early 2025), the EU delays of Meta's multimodal Llama and Apple

Intelligence. The sharpest test came on 12 June 2026, when a US export-control directive forced Anthropic to suspend all access to its frontier Fable 5 and Mythos 5 models for every foreign national overnight — three days after Fable 5's public release. Access was restored 19 days later, on 1 July, after the directive was lifted — but the restoration came from the supplier's side, not from anything the cut-off subjects did, a distinction Part III treats formally (see Evidence Base, Part B).

Autonomy. Can you set your own technology direction — which models you use, how you adapt them, where you take them next? The key question: "Who decides?" This is about power and strategy, and it does not automatically rise or fall with resilience.

The R/A tagging rule. Every sub-indicator in the field guides carries an [R] or [A] tag. The assignment rule: if the sub-indicator measures the ability to *keep operating* under interruption (fallbacks, reserves, diversification, egress control, incident response), it is [R]; if it measures the ability to *set direction* (own models and data, rule-making, indigenous design, talent, governance), it is [A]. Each assessment reports both subtotals. The profile matters as much as the headline: "68 (R 40 / A 28)" and "68 (R 28 / A 40)" call for opposite investment strategies.

1.2 Why a single index will not suffice

A single 100-point scale cannot judge a government and a company at the same time, because the same pillar measures opposite things at the two levels. Take Regulatory: France scores high because it makes the rules — it helped write the EU AI Act and can pass its own national AI laws. BNP Paribas scores high on the same pillar for the opposite reason: following rules well (DORA, GDPR, SR 11-7, ISO/IEC 42001). Each is a sensible measure on its own level; combined, they cancel. Hardware works the same way: a government scores by controlling exports and hosting fabs; a company scores by spreading purchases across suppliers.

The asymmetries run deeper: governments can make money from regulation (the 15% H20 deal) while companies are the ones regulated; governments negotiate chip access while companies receive an allocation; states may own fabs, firms almost never; governments host summits, companies sign codes of practice. Two side-by-side measures make these differences visible; one combined score hides them. This is the framework's core design decision, and it is why the package contains two field guides rather than one.

1.3 Four levels of analysis

Level	Subject	Typical horizon	Primary risk
National	State, EU, regional bloc	10—20 years	Political cut-off; technology denial
Sectoral	Industry (banks, energy, telecoms)	5—10 years	New rules; sector-wide risk
Corporate	Single firm or group	3—5 years	Supplier lock-in; price shocks; compliance failure

Product	Product team, specific project	6—18 months	Service or licence changes; retirement

AI-SAF-C serves the corporate level and, aggregated, the sectoral level; AI-SAF-N the national level. Product-level scoring borrows the vocabulary but its numbers should not be compared between companies.

1.4 The temporal dimension

Sovereignty is a direction of travel, not a fixed position. Someone scoring 55 today but climbing is in better shape than someone at 70 but slipping. An honest assessment needs four reference points: today’s score (T0); the likely score three years out if nothing changes; the likely score three years out under a specific set of planned actions; and speed relative to peers. In a fast-moving field, anyone who stands still is losing ground without doing anything wrong, because the bar for “good enough” keeps rising.

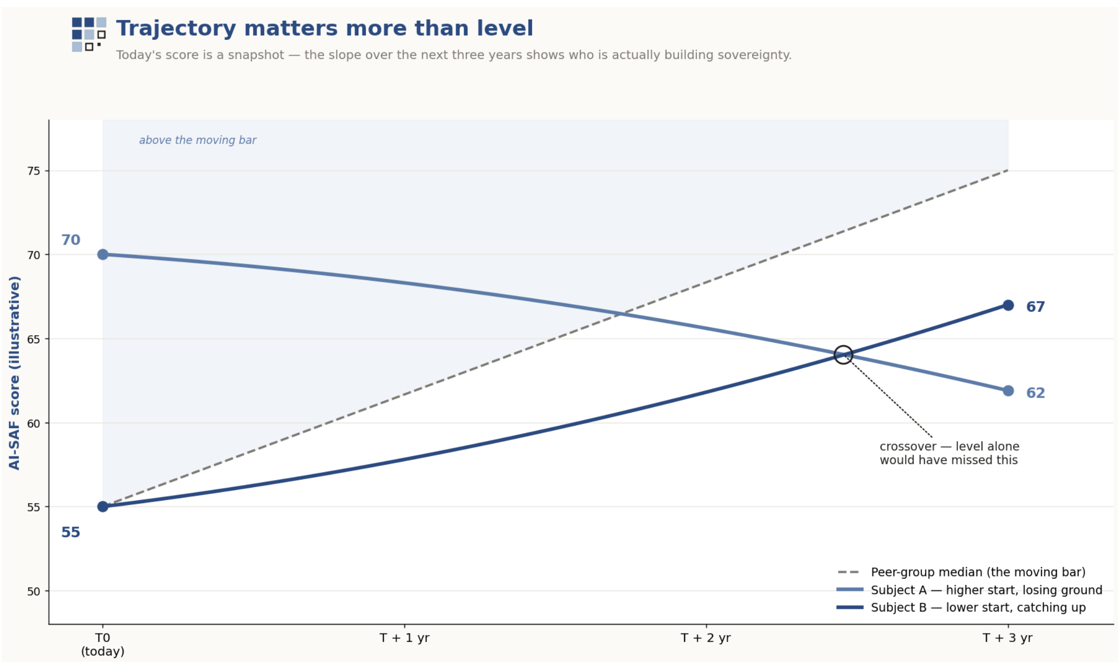


Figure 1.4. An example path from today (T0) to three years out. Subject A starts above the middle of the pack but trends down; Subject B starts below it but rises; the bar itself rises too. Illustrative numbers only.

1.5 Decomposition: capability × commitment

Every sub-score splits into two parts — an idea adapted from the ECFR European Sovereignty Index:

- **Capability** — what is actually in place: training data held, researchers employed, chips installed and running, rules signed and in force, clouds actually certified.
- **Commitment** — what has been promised: strategy papers, announced budgets, laws in progress, agreements signed but not yet in effect.

A country that has promised much but built little is in a very different position from one with the opposite mix. Both numbers are shown for every pillar; the main score uses capability, and commitment drives the three-year projection. A promise with nothing built is just a promise; lots built with no plan is coasting.

1.6 Dependency ratios

Each pillar is also broken down to show exactly where you depend on others, because not every dependence is equally risky:

Dependency axis	Indicator	Reading
Compute	Share of AI computing on foreign-owned platforms	Hard to fix once it passes roughly 70% (author judgement)
Models	Share of top models in use built abroad	Replaceable while open-weights substitutes exist
Semiconductor	Share of chips made abroad	Structural for more than 90% of countries (author judgement)
Software	Share of the core toolkit under foreign control	Replaceable in part
Data	Share of training data from foreign sources	Degrading over time as AI-generated text spreads

Each dependence is labelled a **bottleneck** (replaceable at reasonable cost) or a **structural constraint** (no realistic replacement in sight). The goal is not to remove every dependence — impossible — but to spot the ones you cannot replace and steer spending toward them. The thresholds above are calibration judgements, not published constants; treat them as defaults an assessor may overrule with evidence.

Part II. Pillars and weights

2.1 The eight shared pillars

The backbone is eight pillars, stable against the 2024–2026 incident record and covering the design space of the publicly comparable instruments in active use (Stanford HAI AI Index, OECD.AI, IMF AIPI, Tortoise, CSET/ETO, ECFR, GAIA-X, NIST AI RMF, ISO/IEC 42001).

#	Pillar	Scope
1	Data	Corpus control, linguistic identity, residency, no-train discipline, data egress at inference
2	Models	The L0–L4 ladder from foreign API to from-scratch training; fine-tune capacity; reasoning; portfolio and jurisdiction diversification
3	Infrastructure and energy	Sovereign cloud, AI factories, network and energy independence, confidential computing

4	Hardware architecture	CPU ISA (x86/ARM/RISC-V) and AI accelerators, export-tier access, indigenous designs
5	Manufacturing (foundry)	Leading-edge foundry access, advanced packaging, critical minerals, domestic capacity
6	Software stack	CUDA exposure, training frameworks, inference runtimes, MLOps, open-source contribution
7	Human capital	Researchers at top venues, ML engineers, education pipeline, retention, AI literacy
8	Regulatory and legal	AI-SAF-C: depth of compliance. AI-SAF-N: rule-making capacity

The Data pillar covers not only the data a subject holds but also proprietary, corporate, or personal data that flows out to third-party models at inference time — prompts, retrieved context, fine-tuning sets and logs — scored through data-egress, no-train/data-rights and (for firms) IP-leakage sub-indicators. Uncontrolled egress of proprietary data is now a primary sovereignty exposure; it is cross-referenced to the Models pillar's run-it-yourself fallback rather than counted twice.

2.2 The ninth pillar: Culture and operating model

AI-SAF-C only. Corporate execution does not reduce to technology and compliance. The CrowdStrike incident (19 July 2024: 8.5M Windows machines down, Fortune 500 losses above $5B) showed that the gap between hours and days of recovery was determined by runbook and incident-response maturity, not cloud choice. Every comparable business framework (McKinsey, BCG, Gartner, Deloitte, MIT CISR) treats culture as a distinct dimension. The ninth pillar measures board oversight, applied AI literacy, deployment speed, model-change governance, and disciplined experimentation. States do not have an organisational culture in this sense; the pillar does not appear in AI-SAF-N.

2.3 Weights by variant

Weights reflect the asymmetry between firm and state. The two largest movements — Manufacturing (1 corporate, 12 national) and Culture (10 corporate, absent national) — are mirror images: firms live or die by how they operate; states live or die by fab capacity.

Pillar	AI-SAF-C	AI-SAF-N	Weight rationale
1. Data	14	10	C: corpus + residency + contractual and egress discipline; N: residency emphasis
2. Models	16	16	Portfolio and jurisdiction spread, fine-tune, open fallback, reasoning
3. Infrastructure	14	17	Sovereign cloud, AI factories, CLOUD Act exposure
4. Hardware	10	17	C: reserved GPU, diversification; N: access + design

5. Manufacturing	1	12	C: awareness only; N: fabs, packaging, minerals
6. Software	10	8	CUDA exposure, MLOps, FOSS share
7. Human capital	13	12	C: AI-literate engineer ratio; N: talent stack + retention
8. Regulatory	12	8	C: compliance depth; N: rule-making
9. Culture	10	—	Corporate only
Total	**100**	**100**	

Models keeps 16 in both variants because the 2024–2026 record showed model access as equivalent to ownership. The June 2026 Fable 5 / Mythos 5 suspension settled the point after the weights were set: enterprises built on Anthropic's frontier models lost them outright by government directive, with no ownership to fall back on. That access was restored on 1 July does not weaken the case — a 19-day outage imposed by a foreign government, ended at that government's discretion, is precisely the exposure the 16-point weight prices in (Evidence Base, Part B). The weights are stress-tested rather than taken on faith — see §4.3 for what has and has not yet been run.

2.4 Interpreting the headline score

Band	Level	Meaning
85–100	Sovereign hegemon	Sets the rules; others depend on it
65–84	Strategically autonomous	Operates under interruption; sets its own trajectory
45–64	Partially autonomous	Resilient to some shocks, vulnerable to others
25–44	Dependent with reserves	Critically dependent but has partial alternatives
0–24	Fully dependent	No sovereignty; strategic vulnerability

Calibration notes. Small states (<10M population) cannot realistically exceed ~70 on AI-SAF-N because Manufacturing is structurally out of reach; enforceable participation in pooled EU compute (AI Factories / Gigafactories) partially rebuts this on the Infrastructure side. Every EU bank scores ≥35 on compliance depth alone; for mid-cap EU banks the 35–45 range rebases to 45–60. The AI-SAF-C scale discount is defined in the Corporate Field Guide, Part III.

Part III. The Version Deadlock framework

Version Deadlock is the trap where you can no longer keep your technology up to date, because access to the next version of something essential has been cut off. Depending on a supplier here and now can be managed with contracts, stockpiles, and switching plans; losing the ability to keep up is slow, hard to spot, and often impossible to undo. It happens to countries and companies alike.

3.1 The four types

Type	Concrete scenario	Early-warning indicators

Model	Losing access to the next version of a core model, or to the model itself: the Fable 5 / Mythos 5 suspension (June 2026); EU hold-ups of Meta's multimodal Llama (2024) and Apple Intelligence (2024—25); the DeepSeek bans (2025)	Retirement notices; versions blocked in certain countries; price rises faster than inflation
Hardware	Export limits on newer chips: the Biden Diffusion Rule (Jan 2025, scrapped May 2025); the H20 15% and H200 25% revenue-share deals (2025); case-by-case licence review (2026)	Export-rule changes; sanctions; supply chains redrawn; one-off deals
Software	CUDA licensing changes; key components moved behind closed doors; governance shifts. The CrowdStrike outage (July 2024) was an accidental test of over-reliance on one piece of software	Licence changes; features gated to big customers; single-supplier reliance
Data	A supplier cutting off training data or a licensed dataset; rulings such as NYT v. OpenAI (ongoing), Bartz v. Anthropic (fair use 2025; $1.5B settlement approved July 2026), Kadrey v. Meta (2025), Getty v. Stability (UK, 2025)	Stricter renewal terms; terms-of-service changes; residency rules

3.2 Common mitigation strategies

Model: keep the last working version on your own systems; adapt an open-weights model (Llama 4, Gemma 3, Qwen 3, DeepSeek) as a live fallback; contract for minimum support periods; spread across providers *and jurisdictions* — two or three top providers plus your own open copy in-house. **Hardware:** buy from several suppliers; keep reserves; qualify alternative accelerators before you need them; use confidential computing to run on others' clouds with your data sealed. **Software:** favour open-source tools under multi-party governance (PyTorch Foundation); contribute back; use source-code escrow; keep vLLM, SGLang, llama.cpp as inference fallbacks. **Data:** build your own collection pipeline; gather text in your own language; use synthetic data only as a supplement (models trained mostly on it degrade — Shumailov et al., Nature 2024); bar suppliers from training on your data; align contracts with EU Data Act Chapter VII.

3.3 Version-Deadlock-related modifiers

Rule-stability. The defining feature of 2025—2026 is one-off deals struck case by case, where the rules can change at any time in either direction: exports banned, then taxed at 15%, then licensed; a frontier model suspended by directive, then restored 19 days later. The modifier prices exposure to this volatility — not the strictness of any one rule — and lowers the Hardware and Models scores of anyone at the mercy of unpredictable decisions in a supplier jurisdiction (×0.85 high exposure — ×1.00 none). It applies as a modifier in AI-SAF-N; AI-SAF-C captures the same risk inside its Models pillar through jurisdiction diversification.

Geopolitical-alliance. AI-SAF-N only. The closest US allies get better chip access. That access is not a strength they built, so it multiplies the relevant scores (×0.70–1.15 on Hardware, Models, Infrastructure) rather than adding points. Note that the alliance multiplier and the rule-stability modifier pull against each other for close US allies — alignment buys access and simultaneously deepens exposure to the supplier's rule volatility. Both must always be applied; the National Field Guide fixes the order.

3.4 Historical analogues

Three cases from outside AI set up the categories, and one from AI itself now completes them.

Windows XP in business computing (escaped). From roughly 2007 to 2014, companies kept running XP long after the security risks said to move, because their key business programs worked on XP but not Vista. The real upgrade cost was rewriting those programs. The deadlock was broken from the supplier's side: Microsoft released Windows 7 with an XP compatibility mode. The lesson: some Version Deadlocks are ended by the supplier — and waiting for that is a gamble that their interests stay aligned with yours.

Russian-designed nuclear reactor fuel (slow-clock). VVER reactors across Eastern Europe use fuel rods shaped for that design. Since 2022, operators have been moving away from Rosatom; alternatives exist (Westinghouse), but qualification for a given reactor takes years. The lesson: an alternative that exists on paper but needs five to ten years to qualify is not an alternative within the time-frame that matters.

Soviet industrial licensing (fatal). From the 1960s the USSR acquired Western industrial and computing technology by licence (Tolyatti, the ES EVM computers copying IBM designs). By the late 1970s, COCOM restrictions tightened access to the next generation — and decades of leaning on licensed designs had hollowed out the home-grown ability to design from scratch. The outside limit and the inside decay arrived together. The most important lesson of the three: fatal deadlocks are rarely caused by the moment access is cut; they are caused by the long, comfortable stretch beforehand, when the ability to build your own was allowed to waste away.

Fable 5 / Mythos 5 — the first live AI case (June–July 2026). On 12 June 2026 a US export-control directive forced Anthropic to suspend all access to its frontier Fable 5 and Mythos 5 models for every foreign national, overnight, three days after Fable 5's public release. On 30 June the directive was lifted; Fable 5 returned globally on 1 July, while Mythos 5 was reintroduced only to approved US organisations. The episode compresses the analogues' lessons into 19 days. For subjects that had kept a fine-tuned open model in reserve, it was a Recoverable event; for those that had not, it was three weeks of dead stop and would have been Slow-clock had the directive held. And like Windows XP, the ending came from the supplier's side — the US government reversed itself; no cut-off subject restored its own access. It is the clearest warning the framework has had that a frontier model can be switched off by the jurisdiction that hosts it, that the switch can be thrown and un-thrown inside a month, and that the comfortable years of pure API dependence are exactly when the in-house ability to switch must be paid for.

3.5 Taxonomy of Version Deadlock states

Two questions build the grid. Does another supplier of the version you need exist? Have you kept your own ability to switch, or has it wasted away?

- **Recoverable** — an alternative exists and you have kept your ability to switch. Migration is doable at reasonable cost and time. Windows XP → 7 sits here.
- **Slow-clock** — an alternative exists, but your ability to switch has wasted away; rebuilding the skills to qualify and integrate it takes years. Russian reactor fuel sits here.
- **Vulnerable but recoverable** — no alternative exists, but you have kept the ability to build one, at real cost. Whether you do comes down to political will.
- **Terminal** — no alternative, and no ability left to build one; the gap widens faster than you can close it. Soviet computing by the mid-1980s. The only useful policy is never to land here.

Restoration is not recovery. Two of the cases above — Windows XP and Fable/Mythos — ended because the *supplier side* reversed course. The taxonomy classifies what the subject can do; a deadlock ended by supplier grace leaves the subject's own position exactly where it was. Scoring a subject "Recoverable" because a directive was lifted is an assessment error: the correct reading is the state the subject would have been in had the restriction held. Assessors should record supplier-side restorations as evidence about the *supplier's* rule-stability, not about the subject's resilience.

3.6 Detection: leading vs lagging indicators

The visible signs — a licence refused, retirement notices, prices that shut you out — show up late; by then the deadlock has set in. The early signs are harder: how many generations behind the edge you are and whether the gap is widening; whether your own ability to switch is growing or wasting; supplier signals about its plans. Your own switching ability is the hardest to measure and the most important; rough proxies include researchers active in the relevant area, alternative suppliers qualified in the last cycle, and how long qualification takes when you try. Reaction speed must be tuned to the field: the AI edge moves in months, business software in years, nuclear fuel in decades.

3.7 Policy implications

The point is not whether you depend on others — almost everyone does, for almost everything — but whether you can keep improving while you depend on them. Cutting ties completely is rarely possible. Spreading across suppliers buys time but does not solve the problem. Keeping your own ability to switch is the overlooked lever: the single most important choice during the easy years is to pay, at real expense, to maintain the home-grown ability you would need if access ended. That upkeep wins no political credit, because it produces nothing visible exactly when it matters most — and, going by the Soviet case, it is the difference between a deadlock you can escape and one you cannot. The EU's AI Factories and Gigafactories programmes, national open-model efforts (Bulgaria's BgGPT and peers), and certified sovereign cloud are all best understood as insurance against a future Version Deadlock. The full account is in the companion paper "Version Deadlock: Technological Evolution Under Constrained Dependency" (April 2026).

Part IV. Gap-analysis methodology

The framework's most useful job is not the score but pointing to where the next investment does the most good: "Given where you stand, what you can spend, and how long you have, which pillar gives the biggest gain per unit of money or effort?"

4.1 A four-step process

1. **Score today** — fill in the scale with hard evidence, splitting capability from commitment; note the direction of travel.
2. **Project three years out at current effort** — the bar rises, so standing still usually means falling behind.
3. **Work out payoff per pillar** — points gained per standard chunk of investment (€1M, 10 FTE, or 12 months); pick the pillars that pay most and fastest.
4. **Choose the mix** — the set of moves giving the best three-year position for the budget, counting knock-on effects (people multiply every other pillar's payoff).

4.2 Typical priorities

Companies: first people (the multiplier), then L2 on Models (adapting an open model — realistic, big payoff), then software stack and data. Manufacturing and Hardware take country-sized investment to move. **Countries:** infrastructure and people pay most but need 5–10 years of steady commitment; Models and Regulatory can move faster but hit lower ceilings.

4.3 Methodological rigour — design and current status

The framework is designed to the standard for composite indicators set out in the OECD/JRC Handbook on Constructing Composite Indicators: min-max normalisation to a common scale; imputation only where no more than 15% of data is missing; principal-component cross-checks on the aggregate; and Monte Carlo sensitivity runs that nudge every weight to test whether profile readings — and the ordering of any two subjects more than a few points apart — survive.

Status, stated plainly: as of v0.7.1 these checks are a design commitment, not a completed exercise. The sensitivity runs and PCA cross-checks have not yet been executed and published; they are a precondition for a 1.0 release, alongside the reproducibility test below. Until then, the worked examples should be read as calibrated illustrations, not validated measurements.

Reproducibility between assessors. Where a sub-indicator calls for qualitative judgement, two assessors working independently from the same evidence should land within one scoring band; a wider gap is treated as a defect in the rubric, not the assessors. Score sheets record the evidence relied on so a third party can re-derive the number.

Data sources (catalogued in Evidence Base, Parts G and H): CSET/ETO Supply Chain Explorer (Hardware, Manufacturing); the Epoch AI notable-models database via Stanford HAI (Models); the SecNumCloud 3.2 registry, GAIA-X Level 3 labels, and the EU cloud sovereignty instruments (Infrastructure); CulturaX, HPLT v2, FineWeb-2, INCLUDE, and Global-MMLU (Data); CSRankings, top-venue author counts, MacroPolo Global AI Talent Tracker 2.0, and the Stanford AI Index (Human capital).

Part V. Areas of genuine uncertainty

Every framework has a point past which its own advice turns into guesswork. Being upfront about where that point lies is part of doing the job honestly.

How the AI Act's split timeline plays out. Partially resolved since the last revision: GPAI obligations and Article 50 transparency duties became enforceable on schedule on 2 August 2026, while the Digital Omnibus on AI (in force 27 July 2026) deferred high-risk obligations to December 2027 (Annex III) and August 2028 (Annex I). The open question is now enforcement practice, not the calendar — and how the deferral feeds the three-year commitment projections in AI-SAF-C Regulatory.

Where the Cloud and AI Development Act lands. The Commission's CADA proposal (3 June 2026) introduces a four-level Cloud Sovereignty Framework for public procurement, effectively superseding the stalled EUCS debate. If it survives trilogue near its proposed form, it becomes the EU-wide anchor for the Infrastructure pillar, displacing SecNumCloud as the reference ladder; if it is diluted, SecNumCloud 3.2 stays the top bar. Assessors should track the trilogue.

Whether the big cloud companies' own chips loosen NVIDIA's grip. If TrendForce's forecast (NVIDIA's inference share falling to 20—30% by 2028) is right, CUDA lock-in should matter less in the Software pillar. No decisive evidence either way yet.

Whether export controls on model access become an instrument. The Fable 5 / Mythos 5 episode ended after 19 days, but the precedent stands: the US demonstrated it can and will switch off a frontier model for all foreign nationals by directive. Whether this becomes a repeated tool or stays a one-off, and how far its extraterritorial reach holds up, bears directly on the rule-stability modifier and the Models weight. The DeepSeek V4 question, by contrast, is resolved: V4 shipped with day-0 adaptation to Huawei Ascend (April 2026), confirming that a top Chinese model no longer needs CUDA — advantage China over Europe on the Software pillar.

Life after Magdeburg. Intel's July 2025 cancellation cost the EU Chips Act credibility; Chips Act 2.0 (proposed 3 June 2026) is the policy response, and ESMC Dresden is proceeding. Whether proposal turns into capacity affects Germany's Manufacturing score — a capability-vs-commitment test case.

How much announced money is actually spent. The headline numbers (Stargate $500B, France €109B, the €30B Gigafactories initiative) mix private pledges, partner-country money, and multi-year spending. The framework treats them as rough sizes and separately tracks disbursement.

The framework's own additive form. Pillar scores sum linearly, so a strong Regulatory score can arithmetically offset a near-zero Models score. The framework's own thesis is non-compensatory — sovereignty fails at the choke point. The additive headline therefore overstates sovereignty for unbalanced profiles; the dependency-type labels and the R/A profile are the non-compensatory reading, and assessors should flag any profile where a structural-constraint pillar scores under a quarter of its weight, whatever the headline says. A future revision may adopt partially non-compensatory aggregation; the trade-off is transparency.

Conclusion

AI sovereignty is not about owning the whole technology chain. It is about being able to keep running — and keep improving — if the chain is broken. Every major event of 2024–2026 tested exactly that: the CLOUD Act admission, the H20 and H200 revenue deals, the DeepSeek wave, the Meta and Apple delays, Magdeburg, China's mineral controls, the sovereign-cloud responses — and, above all, the June 2026 suspension of Fable 5 and Mythos 5, the first time a government switched off access to a frontier model itself. That it was switched back on 19 days later, by the same government, only sharpens the lesson: access is not ownership, and restoration is not recovery.

The framework's biggest departure from existing public tools is admitting that the same pillar name means opposite things for a company and a country, and keeping two variants instead of forcing one score. The second is splitting capability from commitment inside every pillar. The third, new in this revision, is carrying the resilience/autonomy distinction into the arithmetic. A business uses AI-SAF-C (Document 2); a government body uses AI-SAF-N (Document 3). This document stays the reference for the method, the terms, and the thinking behind both.

Appendix: Master glossary

AI-SAF / AI-SAF-C / AI-SAF-N: The AI Strategic Autonomy Framework and its business and national variants; eight shared pillars at different weights; AI-SAF-C adds a ninth (Culture).

Base model: A large language model trained from scratch on trillions of tokens (GPT, Claude, Gemini, Gemma, Llama, DeepSeek, Qwen).

Branch-and-Merge: INSAIT's training method (EMNLP 2024) for adapting a model to a new language without catastrophic forgetting.

CADA / Cloud Sovereignty Framework: The EU Cloud and AI Development Act (proposed 3 June 2026); its Cloud Sovereignty Framework grades cloud services on four assurance levels, from EU data location (L1) to full supply-chain control free of third-country interference (L4).

Capability × commitment: Splitting every sub-score into what is actually in place today versus what has been promised.

Chokepoint exposure: Vulnerability at the make-or-break steps of the chip supply chain (EUV lithography, leading-edge logic, HBM, EDA software, key chemicals).

CLOUD Act: The US Clarifying Lawful Overseas Use of Data Act (2018), authorising federal access to data held abroad by US companies.

CUDA: NVIDIA's software layer for its chips; the de facto standard for AI training and the main switching barrier away from NVIDIA.

Data Act Chapter VII: The EU Data Act provisions (fully applicable September 2025) aimed at third-country access.

Data egress: The flow of a subject's own data out to a third-party model at inference time — prompts, retrieved context, tool calls, fine-tuning sets, logs.

Digital Omnibus on AI: The EU amending package (in force 27 July 2026) deferring the AI Act's high-risk obligations to December 2027 / August 2028 while leaving GPAI enforcement on schedule.

DORA: The EU Digital Operational Resilience Act for the financial sector, applicable from January 2025.

FRIA: Fundamental Rights Impact Assessment, AI Act Article 27.

GAIA-X: European certification of cloud and data services against sovereignty criteria at three levels; L3 requires SecNumCloud-grade immunity from foreign-reach laws.

ISA: Instruction Set Architecture — the command set a processor understands (x86, ARM, RISC-V).

ISO/IEC 42001: The certifiable international AI-management standard (December 2023); nine Annex A control groups; a stepping stone to AI Act Articles 53/55.

MoE: Mixture of Experts — a model design activating different specialist sub-models per input.

MRM / SR 11-7: Model Risk Management; the US supervisory standard widely adopted in EU banking — challenger models, monitoring, model inventory, documentation.

NIST AI RMF: The US AI risk-management framework (Govern, Map, Measure, Manage); backbone of the AI-SAF-C crosswalk.

No-train clause: A contract clause barring a model provider from using your inputs or outputs to train its models.

Open-weights: A model whose trained weights can be freely downloaded and run (Llama, Mistral, Gemma, Qwen, DeepSeek); not always fully open-source.

Restoration risk: The error of reading a supplier-side reversal (a lifted directive, an extended support window) as evidence of the subject's own recoverability. See Framework §3.5.

RISC-V: An open, royalty-free ISA; the strategic alternative to x86 and ARM (Tenstorrent, EU DARE, Meta MTIA).

SecNumCloud: France's ANSSI certification (v3.2) — the established scheme requiring a cloud service to be beyond the reach of foreign laws such as the CLOUD Act and FISA.

Strategic interdependence: Deliberately depending on several partners so no single one can dictate — the realistic alternative to autarky for small and mid-sized economies.

Synthetic collapse: The effect shown by Shumailov et al. (Nature, 2024): repeatedly training on AI-generated data permanently strips away rare cases.

TEE: Trusted Execution Environment — hardware-isolated execution (Intel TDX, AMD SEV-SNP, NVIDIA Confidential Computing) enabling private inference on clouds you do not control.

Version Deadlock: The state in which a subject loses the capacity to evolve technologically because access to the next version of a key technology has been interrupted.

vLLM / SGLang / llama.cpp: Open-source inference engines; alternatives to NVIDIA's TensorRT-LLM that run models without CUDA dependence.

Document 2 · AI-SAF-C Field Guide — Business variant

Assessment of corporate AI sovereignty. Version 0.7.1 · August 2026 · CC BY 4.0

About this document

This field guide is the hands-on manual for AI-SAF-C, the business version of the AI Strategic Autonomy Framework, for a CTO's office, a compliance team, or an AI director sizing up how self-reliant their company is and deciding what to fix first. For the concepts behind it — the two variants, capability × commitment, gap analysis, Version Deadlock — see Document 1; this guide assumes them. Dated market detail (which providers are certified, which chips ship) lives in Document 4, State of Play.

How each pillar is presented. Every pillar opens with what it measures and why, then a rubric table: sub-indicator, points, R/A tag, and the evidence expected. Score each sub-indicator as a whole number from 0 to its maximum; the pillar score is the sum. Report R and A subtotals with the headline (Framework §1.1).

Contents: Part I — the nine pillars. Part II — the ninth pillar in depth. Part III — scale discount. Part IV — worked example: BNP Paribas, with an ING calibration check. Part V — crosswalk to NIST AI RMF, ISO/IEC 42001, and GAIA-X.

Part I. The nine pillars

#	Pillar	Weight	Corporate focus
1	Data	14	Own data, where it is stored, contractual and technical control of what leaves
2	Models	16	Portfolio across providers *and jurisdictions*; fine-tune capability; self-run fallback
3	Infrastructure	14	Cloud choice and jurisdiction, CLOUD Act exposure, own GPUs, confidential computing
4	Hardware	10	Reserved GPU capacity; vendor diversification; migration plan
5	Manufacturing	1	Supply-chain awareness only
6	Software	10	CUDA and tooling lock-in; open-source contribution
7	Human capital	13	AI-literate engineer share, retention, enablement programs
8	Regulatory	12	AI Act, DORA, GDPR, ISO 42001, model-risk management
9	Culture and operating model	10	Board oversight, applied literacy, speed, change governance, experimentation
	Total	**100**	

Pillar 1. Data — 14 points

Capable open-weights models have become cheap and widely available, so when everyone can get roughly the same models, your own private data — and the contracts and controls that protect it — becomes your strongest, hardest-to-copy advantage. The pillar scores four things: the corpus you hold, where it legally sits, what a provider may do with what you send, and what actually leaves your boundary at inference time.

No-train and data-rights discipline. The key contract terms: a clause barring the provider from using your inputs or outputs to train or tune models; retention limits (zero to 30 days); processing location (EU-only for sensitive data); named subprocessors; audit rights (at least annual SOC 2 Type II; on-site for regulated industries per DORA Article 30); ownership and deletion of fine-tune artifacts built from your data; and derived telemetry and embeddings under the same terms as the raw data.

Data-egress boundary control. Contracts govern what a provider may do; this governs whether data leaves at all — prompts, retrieved context (RAG), tool and agent calls, logs. **IP and trade-secret exposure** is scored separately: whether high-sensitivity IP (source code, designs, unreleased financials, M&A material) is kept out of, or isolated within, external model calls, with leakage monitoring. **Synthetic data** plays a limited role: fine for padding and test cases, never the core of fine-tuning data (Shumailov et al., Nature 2024 — models trained mostly on model output degrade irreversibly). A **sovereign-inference fallback** —

what share of workloads could run in-house if egress had to stop tomorrow — is scored once, in the Models pillar, and credited here only by cross-reference.

Sub-indicator	Points	R/A	Evidence expected
Own corpus and quality	5	A	Volume and purity of fine-tuning data; domain coverage
Residency and legal regime	3	R	Jurisdiction of storage and processing; CLOUD Act exposure; SCCs
Instruction and evaluation data	1	A	Own preference pairs; domain benchmarks
No-train and data-rights discipline	2	R	Clause quality; audit rights; retention; artifact ownership
Data-egress boundary control	2	R	Classification policy; AI gateway with logging; redaction/DLP; private inference for the most sensitive classes
IP and trade-secret exposure	1	R	Sensitive-IP classes blocked or isolated; leakage monitoring

Pillar 2. Models — 16 points

Three questions: do you spread your bets across models, providers, and jurisdictions; can you fine-tune on your own data; and do you have a model you can run yourself if a Version Deadlock hits.

The five-level ladder, applied corporately (anchors L0—L4 = 0/4/11/13/16):

Level	Points	Corporate profile	Note
L0	0	Entirely on frontier APIs with no fallback	CLOUD Act + Version Deadlock exposure
L1	4	On-prem open-weights fallback, no fine-tune	Physical control only
L2	11	Product-level fine-tune on open weights	Fine-tune capability; open fallback
L3	13	Domain-specific from-scratch models	Rare; only for very specialised needs
L4	16	Foundation-model companies	Not themselves assessed on AI-SAF-C

Most companies should aim for L2. The ladder anchors the level; points are earned through the sub-indicators, which sum to 16, with the anchor as reference.

Portfolio and jurisdiction diversification. The June 2026 Fable 5 / Mythos 5 suspension changed what diversification means. A portfolio of two or three providers all under the same jurisdiction is not diversified against the risk that actually materialised: a directive that in principle reached every customer of every US frontier vendor at once. The sub-indicator therefore scores *legal regimes*, not logos: US-jurisdiction frontier APIs, EU-jurisdiction providers, and self-hosted open weights (which carry no provider jurisdiction at inference) count as distinct classes. A portfolio of three US providers scores at most 2 of 5; provider count within a class adds at most 1 further point. A well-built portfolio runs three layers: a primary frontier service for the highest-quality work; a second provider — ideally in a different jurisdiction — shadow-tested quarterly; and a self-run open-weights model, fine-tuned to your domain, carrying real traffic daily rather than sitting idle. Real resilience degrades gradually rather than stopping dead.

Sub-indicator	Points	R/A	Evidence expected
Portfolio and jurisdiction diversification	5	R	Distinct legal regimes in live use; active traffic split; ≤2 of 5 for single-jurisdiction portfolios
Fine-tune capability	5	A	L2 capability; fine-tuned models in production
Open-weights fallback	4	R	Actively operated on-prem/VPC instance; RTO < 72h; carries live traffic
Reasoning usage	2	A	Reasoning-class models integrated where they fit

Pillar 3. Infrastructure and energy — 14 points

Four questions: which cloud and whose law; CLOUD Act exposure and mitigation; own GPU capacity; confidential-computing readiness.

The jurisdictional ladder for cloud choice runs, from strongest protection down: SecNumCloud 3.2 providers (immune to foreign-reach laws; required for French state clients); the CADA Cloud Sovereignty Framework levels as they are finalised (see Framework Part V); EU-operated subsidiaries of US providers (AWS European Sovereign Cloud — EU-run but still US-owned, so CLOUD Act exposure arguably remains); contractual EU boundaries (Azure EU Data Boundary — Microsoft's own French Senate testimony conceded CLOUD Act protection cannot be guaranteed); and unmitigated US-jurisdiction cloud. **Confidential computing** (TEEs: Intel TDX, AMD SEV-SNP, NVIDIA Confidential Computing) lets a regulated EU company run on a US cloud with weights and inputs encrypted in use — credited as genuine mitigation. Current certified providers and capacity figures: Document 4.

Sub-indicator	Points	R/A	Evidence expected
Cloud choice and jurisdiction	5	R	Certified-sovereign, EU-subsidiary, or documented mitigation strategy
On-prem / in-VPC GPU	4	R	Own or reserved capacity in H100-equivalents
Network and backup independence	2	R	Multi-region, multi-vendor; tested DR runbooks
Confidential-computing readiness	3	R	TEE patterns on regulated workloads; attestation; HSM integration

Pillar 4. Hardware architecture — 10 points

Three questions: how much guaranteed GPU capacity is locked in for a crunch; how reliant on one supplier; is there a tested plan to switch. NVIDIA still dominates, but AMD is a believable second source and cloud ASICs (TPU, Trainium) a third class; details in Document 4.

Sub-indicator	Points	R/A	Evidence expected
Reserved GPU capacity	4	R	Multi-year reservations in H100-equivalents; T+3 projection
Vendor diversification	4	R	Production traffic on at least two accelerator vendors
Migration plan under Version Deadlock	2	R	Documented, tested fallback under export-control scenarios

Pillar 5. Manufacturing — 1 point

A company that does not build foundation models cannot meaningfully score here. One point [R] is given for demonstrated supply-chain awareness — knowing where the packaging and memory bottlenecks are and feeding that into buying decisions. The single point is kept as a separate pillar so the eight shared pillars stay one-to-one with the national variant.

Pillar 6. Software stack — 10 points

Worth two points more than in the national variant because companies feel lock-in in everyday work: CUDA exposure, AI-ops tooling, and inference-runtime choice each directly affect Version Deadlock risk. The runtime landscape and CUDA-alternative status: Document 4.

Sub-indicator	Points	R/A	Evidence expected
CUDA exposure	3	R	Share of production workloads dependent on CUDA alone
Training-framework diversification	2	R	PyTorch + JAX or alternatives; reproducibility discipline
Inference-runtime diversification	2	R	vLLM / SGLang / llama.cpp alongside or instead of TensorRT
MLOps lock-in	2	R	Vendor-agnostic deployment; portability demonstrated
Open-source contribution	1	A	Upstream commits, not merely consumption

Pillar 7. Human capital — 13 points

The pillar with the biggest knock-on effect: gains here improve the payoff from every other investment. BCG's survey work suggests employee training is the single best predictor of AI-program success (strong-results share rising from 12% to 28%).

Sub-indicator	Points	R/A	Evidence expected
AI-literate engineer ratio	4	A	Share of engineers actively working with AI tooling
ML/AI specialists	3	A	Senior ML engineers and researchers on staff
AI Champions / enablement program	3	A	Formal cross-functional program (the AI Factory pattern)
Retention against brain drain	2	R	AI-staff retention metrics; competitive compensation
AI literacy in non-tech staff	1	A	Formal training coverage; internal-assistant adoption

Pillar 8. Regulatory (compliance) — 12 points

For businesses this pillar measures how thoroughly you follow the rules — the mirror image of the national variant (Framework §1.2). The reference obligations: the EU AI Act (GPAI obligations and Article 50 transparency enforceable since 2 August 2026; high-risk obligations deferred by the Digital Omnibus to December 2027 for Annex III systems and August 2028 for Annex I products — deadlines that feed the commitment projection, not an excuse to stop preparing); DORA (January 2025) for financial firms; GDPR plus Data Act Chapter VII (September 2025) for residency and CLOUD Act shielding; SR 11-7-style model-risk management; and ISO/IEC 42001 certification, credited as proof of disciplined procedure.

Sub-indicator	Points	R/A	Evidence expected
AI Act conformity posture	4	A	Risk classification complete; technical documentation; FRIA; registration
DPIA + GDPR + Data Act compliance	2	A	DPIA completeness; lawful basis; SCCs; Chapter VII compatibility
MRM (SR 11-7 equivalent)	3	R	Model inventory; challenger models; monitoring; independent validation
ISO 42001 certification	2	A	Active certification; SoA completeness; audit evidence
DORA readiness (financial)	1	R	ICT risk management; third-party register; incident reporting

Part II. The ninth pillar — Culture and Operating Model

Getting AI right in a company is not just technology and rule-following. The CrowdStrike incident (July 2024) was the clearest recent test of operational maturity: whether recovery took hours or days came down to playbooks and response teams, not cloud choice. Every major business AI-maturity framework (McKinsey, BCG, Gartner, Deloitte, MIT CISR) treats culture as its own dimension, and for the same reason AI-SAF-C gives it a 10-point pillar. "Culture" here means how the company actually operates — measured by what people visibly do (real AI use in everyday work, deployment speed, escalation, behaviour change after incidents), not by mission statements or survey answers.

The ten points split evenly — five for governance, five for practice — on purpose. Governance without practice produces polished policies that people route around; practice without governance produces companies that ship fast and discover accountability only when something breaks.

Governance side (5):

Sub-indicator	Points	R/A	Scoring anchors
AI ethics / model-risk committee	3	A	0 = none → 3 = board-level chair; monthly or on-demand; tied into risk and audit; full risk-ranked model inventory
Written model-risk management	2	A	0 = no policy → 2 = documented checks per live model, monitoring, retirement schedule, staffed independent validation

Practice side (5):

Sub-indicator	Points	R/A	Scoring anchors
Incident handling	2	R	0 = no AI-specific process → 2 = rehearsed response with drills, tested rollback, routine near-miss reporting
Rollout speed	1	R	0 = >6 months or untracked → 1 = ≤3 months, tracked against a target
Disciplined experimenting	2	A	0 = no systematic testing → 2 = own-domain benchmarks; A/B tooling wired into rollout; results gate go-live

Calibration: a long-established, heavily regulated firm with strong governance but regulation-limited speed scores around 8; a mature AI-first scaleup with strong practice but lighter governance around 7; an early-stage firm with neither, 1–2. The even split keeps either type from winning the pillar on company type alone.

Part III. Scale discount

Size affects how realistically some scores can be reached. For enterprises under 500 employees, the Data, Human Capital, and Culture raw scores are multiplied by 0.85; from

500 to 2,000 employees, by 0.92; above 2,000, no discount. The other six pillars are not adjusted — they depend on architectural and contractual choices, not headcount. A 30-person startup cannot build a bank-sized fine-tuning corpus however well it manages data, and formal governance comes more easily at scale; but a small firm can reach L2 on Models, run on certified sovereign infrastructure, hold ISO 42001, and diversify hardware — none of that needs a thousand people.

Part IV. Worked example: BNP Paribas

The strongest example of a European bank doing AI well: a multi-year Mistral partnership since February 2024; an in-house "LLM as a service" platform on BNP's own GPUs in its own data centres; disciplined governance. (Both profiles below are the authors' estimates from public information — see the Evidence Base's verification note.)

4.1 Base profile

Pillar	Weight	Raw	Score	Note
Data	14	0.70	9.8	Large proprietary financial datasets; strict no-train contracts
Models	16	0.55	8.8	L2: Mistral partnership + internal fine-tunes; genuine jurisdiction spread (EU provider + self-run weights); no own FM
Infrastructure	14	0.70	9.8	LLM-aaS on own data centres; backer of Mistral's Essonne DC
Hardware	10	0.35	3.5	Some H100/H200 in-house; NVIDIA-only
Manufacturing	1	0.00	0.0	Not applicable
Software	10	0.55	5.5	Mature MLOps; open weights + proprietary mix
Human capital	13	0.65	8.5	AI Factory / AI Champion programs; large quant population
Regulatory	12	0.85	10.2	DORA, AI Act readiness, GDPR-native, SR 11-7-equivalent MRM
Culture	10	0.55	5.5	Board-level oversight; speed limited by risk committees
Total	**100**		**~61.6**	No scale discount (>2,000 employees)

4.2 Reading the profile

Regulatory is the strongest pillar — banking is the most heavily regulated industry there is, and BNP has two centuries of practice at it. Data is second: huge private financial datasets that no non-bank could assemble, protected by tight no-train contracts. Infrastructure scores on the on-prem platform and the unusual depth of the Mistral relationship — BNP is at once customer and financier. Models has growth room: L2 is the right level for a bank (building a foundation model is not a bank's job), and BNP's portfolio is one of the few in EU banking with real jurisdictional spread — an EU frontier provider plus self-run open weights

— which is what the diversification sub-indicator now rewards. Hardware and Culture are the constraints: NVIDIA dependence is industry-wide, and a big regulated bank deploys slowly by design.

4.3 Calibration comparison: ING Bank

ING made a different choice: primarily Azure OpenAI, less in its own data centres, more CLOUD Act exposure — and, in the revised Models terms, a single-jurisdiction portfolio.

Pillar	Weight	Raw	Score	Note
Data	14	0.55	7.7	Strong datasets; contractual discipline limited
Models	16	0.35	5.6	Primarily Azure OpenAI; single jurisdiction; limited fine-tune; insufficient fallback
Infrastructure	14	0.40	5.6	Azure EU Data Boundary; CLOUD Act exposure; limited on-prem
Hardware	10	0.20	2.0	Minimal in-house; fully cloud dependent
Manufacturing	1	0.00	0.0	N/A
Software	10	0.50	5.0	Standard enterprise stack
Human capital	13	0.55	7.2	Smaller AI staff than BNP
Regulatory	12	0.75	9.0	DORA native; AI Act readiness in progress
Culture	10	0.50	5.0	Moderately slower pace
Total	**100**		**~47.1**	

ING scores about 47, fifteen points below BNP, and the gap sits exactly where the June 2026 episode said it would: Models, Infrastructure, and Hardware. Both are reasonable strategies; AI-SAF-C makes the sovereignty difference visible. During the 19-day Fable/Mythos suspension, a BNP-profile institution would have degraded gradually onto its EU provider and self-run models; an ING-profile institution built on a single US stack had no equivalent cushion — the point of the jurisdiction-diversification revision.

4.4 A structural finding for EU banking

Every EU bank scores at least 35 on compliance depth alone — DORA, the AI Act, GDPR, and MRM; a bank below 35 will draw supervisory action. For mid-sized EU banks the practical range is 45–60. Above 60 requires either a deep partnership with an EU model company or a large self-run model estate — today BNP, potentially Santander, HSBC, or UBS with the same investment. Above 70 would require building a top-tier model from scratch, which is not a realistic goal for a bank.

Part V. Crosswalk to business standards

AI-SAF-C is not a separate audit. It is a higher-level score that reuses evidence you already have — NIST AI RMF playbooks, ISO/IEC 42001 Statements of Applicability, GAIA-X selfdescriptions — turning “yet another assessment” into a reuse of compliance work already paid for.

AI-SAF-C pillar	NIST AI RMF	ISO 42001 Annex A	GAIA-X criterion
Data (14)	MAP 2.3, MAP 3, MEASURE 2.8/2.10	A.7 Data + A.4	Data protection; Portability; European control
Models (16)	MEASURE (all) + MANAGE 4.1; GenAI Profile	A.6 Lifecycle + A.5	Transparency
Infrastructure (14)	GOVERN 6, MAP 4, MANAGE 3	A.4 Resources + A.10 Third-party	Security + European control (L2/L3)
Hardware (10)	GOVERN 6.1	A.4 + A.10	European control
Manufacturing (1)	—	—	—
Software (10)	GOVERN 1.5, MEASURE 2.7, MANAGE 2.3	A.6 + A.10	Security (SDL controls)
Human capital (13)	GOVERN 2, 3, 4	A.3 + A.4.5	Transparency (partial)
Regulatory (12)	GOVERN 1 + MAP 1	A.2 + A.5 + A.8	Contractual governance + Data protection
Culture (10)	GOVERN 3	A.3 Internal organisation	—

How to use it: gather the compliance paperwork you already hold; pull the matching evidence per pillar via the table; answer the few questions the paperwork does not cover (mainly: a workable self-run fallback, no-train discipline, deployment and experimentation speed); report the AI-SAF-C score alongside the existing compliance reports, not as a separate document. For EU companies serving regulated customers, GAIA-X Level 3 feeds straight into the Infrastructure score; Level 2 is a sensible minimum for business software sold to non-regulated customers. As the CADA Cloud Sovereignty Framework is finalised, its assurance levels will slot into the same column.

For the master glossary, see Document 1. Terms used here without definition are defined there.

Document 3 · AI-SAF-N Field Guide — National variant

Assessment of national AI sovereignty. Version 0.7.1 · August 2026 · CC BY 4.0

About this document

This field guide is the hands-on manual for AI-SAF-N, the national version of the AI Strategic Autonomy Framework, for a government body, ministry, or regional bloc measuring its AI sovereignty and building strategy from the result. For the concepts — the two variants, capability × commitment, gap analysis, Version Deadlock — see Document 1. Dated market detail lives in Document 4, State of Play.

How each pillar is presented. Each pillar opens with what it measures and why, then a rubric: sub-indicator, points, R/A tag, evidence expected. Whole-number scores; the pillar score is the sum; report R and A subtotals with the headline.

Contents: Part I — the eight pillars. Part II — the four modifiers and their fixed order of application. Part III — worked examples: Bulgaria, France, UAE. Part IV — case study: INSAIT and BgGPT.

Part I. The eight pillars

#	Pillar	Weight	National focus
1	Data	10	National corpus, residency, legal regime, public-sector egress
2	Models	16	The L0—L4 ladder; reasoning bonus
3	Infrastructure and energy	17	Sovereign cloud, local and pooled compute, grid
4	Hardware architecture	17	Export tier, indigenous design, supplier spread
5	Manufacturing	12	Fabs, advanced packaging, critical minerals
6	Software stack	8	CUDA dependence, framework diversity, FOSS contribution
7	Human capital	12	Top-venue researchers, retention, pipeline
8	Regulatory (participation)	8	Rule-making seats, own binding law, institutions, summits
	Total	**100**	

Pillar 1. Data — 10 points

For languages with less material online, data limits sovereignty more than models do: open-weights models are now cheap and common, while a genuine national corpus is not. Combined totals across the big open collections (CulturaX, HPLT v2, FineWeb-2, deduplicated) sort languages into tiers: **Low** (5—40B tokens — Bulgarian, Estonian, Welsh: not enough to train mid-sized models from scratch; continued pre-training is forced), **Medium** (40—250B — Polish, Greek, Vietnamese, Hebrew), **High** (250B—1.5T — German, French, Arabic: from-scratch training feasible). On the legal side, the EU regime (GDPR + Data Act, with Chapter VII targeting third-country access + AI Act) is the world's strictest; India's DPDP rules are gentler; China's PIPL is strictest on government data.

Public-sector data egress complements residency: whether sensitive government and critical-sector data is sent to foreign-controlled models at all, and whether a sovereign-inference option (certified sovereign cloud plus a domestic or open model) exists and is mandated.

Sub-indicator	Points	R/A	Evidence expected
Corpus authenticity	4	A	Volume and cleanliness of local text; share real rather than AI-generated
Residency and legal regime	3	R	Where data sits; governing law; CLOUD Act exposure
Instruction and evaluation data	1	A	Coverage in INCLUDE / Global-MMLU; local instruction data
Public-sector and proprietary-data egress	2	R	0 = no rule; 1 = binding procurement bar on sensitive data to foreign models; 2 = sovereign-inference option mandated for sensitive workloads

Pillar 2. Models — 16 points

The pillar that separates countries most. The five-level ladder (anchors L0—L4 = 0/4/11/13/16):

Level	Points	National example, 2026	Note
L0	0	Governments on foreign frontier APIs	CLOUD Act and directive exposure
L1	4	Open-weights models run on-prem, unadapted	Control of the hardware, nothing more
L2	11	BgGPT (Gemma 3 → Bulgarian); Meltemi; Sarvam-M; TAIDE	Adapt or keep training someone else's base model
L3	13	EuroLLM; Teuken; Poro; Aya-Expanse; EXAONE	Own model and tokeniser, on foreign compute
L4	16	Fugaku-LLM; Falcon/Jais (UAE); DeepSeek/Qwen on Ascend	Fully independent, top to bottom

Scoring rule: the ladder anchors the level; the pillar score is the assessor's raw fraction × 16, with the anchor as the reference point. Worked examples show both (e.g. "L2 anchor 11; raw 0.55 after portfolio-depth deductions"). The jump from L2 to L3 turns on owning the tokeniser — the part that splits language into pieces — which removes a built-in bias toward foreign languages that is even harder to fix than the model itself.

Reasoning bonus: a country that has released a reasoning model in its own national language earns +2 within the pillar, capped at 16. Currently: China, South Korea, India, UAE.

Spending thresholds: DeepSeek-V3's reported ~$5.6M in rented compute sets the floor for replicating a top model; roughly $500M/year is the bar for matching a full national ecosystem.

Sub-indicator	Points	R/A	Evidence expected
Ladder level (anchored)	12	A	Highest level achieved, evidenced by shipped models in use
Portfolio depth and open fallback	4	R	More than one usable model; self-hosted deployment in government

Pillar 3. Infrastructure and energy — 17 points

Every major sovereignty incident of 2024—2026 ran through this pillar. The scoring anchors for sovereign cloud, in descending order of protection: **SecNumCloud 3.2** (immune from foreign-reach law — the established top bar); the **CADA Cloud Sovereignty Framework** levels once finalised (the proposed EU-wide four-level ladder — track the trilogue; see Framework Part V); **GAIA-X Level 3**; EU-operated subsidiaries of foreign providers (BSI C5 tier — the AWS European Sovereign Cloud sits here: EU-run, still US-owned). Do not score against EUCS: its sovereignty tier was removed in March 2024 and never returned; CADA is where that debate moved.

Local compute credits AI-accelerator capacity on the subject's own territory at full rate, plus up to 2 of its 4 points for *enforceable* pooled EU capacity — a signed EuroHPC AI Factory or Gigafactory hosting or consortium agreement conferring allocation rights the subject can invoke. The Gigafactories call opened on 30 July 2026 (seven sites, €10B public funding, awards early 2027): a signed award will qualify; a bid does not. Mere programme membership scores 0. Pooled capacity is labelled a bottleneck, never a structural asset.

Sub-indicator	Points	R/A	Evidence expected
Sovereign cloud / cloud choice	5	R	Certification against the anchors above; European jurisdiction
Local compute	4	R	Own-territory accelerator capacity; enforceable pooled allocation (≤2)
Network independence	2	R	Several providers plus own interconnects
Energy mix and price	3	R	Diversified sources; competitive price; grid headroom
Cryptographic control + TEE	3	R	HSMs, key management, attested confidential computing

Pillar 4. Hardware architecture — 17 points

The fastest-changing pillar: US export rules went through seven major turns between 2022 and 2026, and the 2026 watchword is deal-by-deal diplomacy — bans became revenue-share deals became case-by-case licences. Vendor and alternative-chip detail: Document 4.

Sub-indicator	Points	R/A	Evidence expected
CPU ISA diversification	5	R	Mix of x86 / ARM / RISC-V; home-grown designs (SiPearl, DARE)
AI accelerators — access	7	R	Export tier; reserved capacity; supplier spread
AI accelerators — indigenous design	5	A	Realistic only at EU scale; Rhea-, EXAONE-, or TPU-class effort

Pillar 5. Manufacturing — 12 points

The most lopsided pillar: only Taiwan, South Korea, the US, and (to a limited degree) China can make advanced chips in volume; for everyone else this is the dependency that cannot be reduced, only diversified. Current fab, packaging, and HBM state of play: Document 4. **Critical minerals** are scored inside this pillar (2 points): the share of gallium, germanium, heavy rare earths, high-purity silicon, and refined copper a country (or its allies) can refine at home. Ore in the ground does not count — refining is the chokepoint, as China's 2024–2026 export-control campaign demonstrates.

Sub-indicator	Points	R/A	Evidence expected
Supplier diversification	4	R	Access to at least two foundry partners
Local advanced packaging	3	R	Chiplet assembly, test, packaging at national or EU level
Own foundry capacity	3	A	A real fab under own control
Critical minerals	2	R	Domestic/allied refining share of Ga, Ge, rare earths, Si, Cu

Pillar 6. Software stack — 8 points

CUDA still dominates but its grip is loosening; multi-party-governed open source (the PyTorch Foundation model) is much harder for any one company to weaponise than single-vendor software — a modest structural plus for the EU. Runtime and framework state of play: Document 4.

Sub-indicator	Points	R/A	Evidence expected
CUDA exposure	2	R	Share of national AI workloads that run only on CUDA
Training-framework diversification	2	R	PyTorch, JAX, and alternatives in real use
Inference-runtime diversification	2	R	vLLM / SGLang / llama.cpp deployed
MLOps lock-in	1	R	Vendor-agnostic deployment patterns
Open-source contribution	1	A	Upstream contributions from national institutions

Pillar 7. Human capital — 12 points

The ultimate bottleneck. Reference sources: the MacroPolo Global AI Talent Tracker 2.0 (March 2024, NeurIPS 2022 data — note the correct edition is 2.0) for talent flows; the Stanford AI Index for the annual picture; CSRankings and top-venue author counts for national standing. Headline 2026 context — US ~75% of global GPU compute, 50 notable US models vs 30 Chinese in 2025, Switzerland and Singapore leading per-capita researcher density — is kept in Document 4 with the other dated figures.

Sub-indicator	Points	R/A	Evidence expected
AI researchers / top-tier	4	A	Top-venue papers; CSRankings standing; talent based at home
ML engineers and operators	3	A	Production experience; qualified workforce size
Education pipeline	2	A	Strong programs and intake volumes
Retention against brain drain	2	R	Pay competitiveness; career paths; track record
AI literacy in the population	1	A	Formal training reach; basic digital skills

Pillar 8. Regulatory (participation) — 8 points

In the national variant this pillar measures the power to make the rules — the mirror image of the corporate variant (Framework §1.2). Reference instruments as of August 2026: the EU AI Act (GPAI obligations enforceable since 2 August 2026; high-risk obligations deferred by the Digital Omnibus to December 2027 / August 2028); the Council of Europe Framework Convention on AI (in force 1 November 2025 — the first binding international AI treaty); the summit series (Paris, February 2025; New Delhi, February 2026, whose Declaration was endorsed by 92 countries and organisations); and the sandbox obligation (at least one AI regulatory sandbox per member state).

Sub-indicator	Points	R/A	Evidence expected
Rule-making seats	3	A	Seats in bodies writing binding rules
Own binding AI law	2	A	Applying the AI Act plus national implementing law
Institutional presence	2	A	An AI safety/security institute; Council of Europe participation
Summit hosting + bilateral frameworks	1	A	Hosting summits; bilateral AI agreements

Part II. Nation-specific modifiers

Four adjustments applied after the pillar scores, reflecting how unstable the rule environment has become. **Apply them in this order, always, and all of them:** (1) rule-stability, (2) geopolitical-alliance, (3) operating-model. (Critical-minerals is scored inside Manufacturing and needs no separate step.) Worked examples must show each step explicitly; no other adjustments — bonuses, penalties, or otherwise — are permitted outside these four. If a strength seems to deserve extra credit (abundant energy, rule-making influence), it belongs inside the relevant pillar's sub-indicators, where it is already counted.

2.1 Rule-stability modifier (×, on Hardware and Models)

The defining feature of 2025–2026 is one-off deals and reversible directives — exports banned, then taxed, then licensed; a frontier model suspended, then restored within a month. The modifier prices exposure to that volatility.

Exposure	Multiplier	Example
None — compute home-grown	1.00	China (partly); countries with own large providers
Low — deliberately multi-supplier	0.95	European countries spreading purchases
Moderate — mostly US-supplied	0.90	Most EU states today
High — case-by-case US permission	0.85	UAE; countries with no alternatives

2.2 Geopolitical-alliance multiplier (×, on Hardware, Models, Infrastructure)

The closest US allies get better chip access. That access is not a built strength, so it multiplies rather than adds. Note the tension: Tier-1 alignment raises this multiplier *and* typically raises rule-stability exposure — the two must both be applied, which is why the order is fixed.

Position	Multiplier	Examples
Tier-1 US ally	1.10–1.15	Japan, UK, Canada, Australia, South Korea, Netherlands, Taiwan, Singapore, France, Germany
Special partnership	1.05–1.10	UAE (US–UAE AI Acceleration Partnership)
Neutral	1.00	Bulgaria, India, most developing nations
Restricted access	0.90–0.95	Below-tier under export rules
Sanctioned	0.70–0.80	Iran, Russia, (partially) China

2.3 Critical-minerals modifier

Scored inside Manufacturing (2 of 12): domestic or allied refining capacity for gallium, germanium, heavy rare earths, high-purity silicon, and refined copper. China's export-control campaign — gallium/germanium (December 2024), rare earths (October 2025, paused November 2025 for 12 months, with active enforcement through 2026) — shows the risk is present, not theoretical.

2.4 Operating-model modifier (integer, −2 to +2, on the headline)

How a government organises itself to build and keep its own AI capability — durability across political change, legitimacy, exit costs, reproducibility, decision clarity. Four common forms: a public research institute (Bulgaria's INSAIT), a state-owned company (several Chinese labs), a private national champion (France's Mistral), a consortium (EuroHPC AI Factories). No form is favoured; any can earn +2 or −2. The modifier does take one side: a real, coherent setup of your own beats renting everything from foreign providers — disagreeing with that is disagreeing with AI-SAF as a whole.

Scoring: **−2** no real setup, or paper-only. **−1** a setup exists but under-funded, badly run, or unclear. **0** solid, funded, documented, producing results used at home; sovereignty as by-product. **+1** all that, with sovereignty built in on purpose (openness, clear decision-making, reproducibility) but untested through a major upheaval. **+2** proven through at least one real test — change of government, funding gap, major shock.

Calibration: Bulgaria/INSAIT +1 (coherent, resourced, explicitly open; untested through political discontinuity). France/Mistral +1 (commercial discipline plus state alignment; also untested). A small state renting everything through foreign hyperscalers: −1 or −2.

Part III. Worked examples

All three examples follow the same sequence: pillar table → rule-stability → alliance → operating-model → final. These are the authors' estimates from public information, shown to demonstrate the method.

3.1 Bulgaria — ~46

Bulgaria punches above its weight: a small economy (~$100B) whose INSAIT institute built far more national capability than size suggests, by continuing to train Google's Gemma models into BgGPT and putting the result into government use (7,000+ staff at the National Revenue Agency).

Pillar	Weight	Raw	Score	Note
Data	10	0.70	7.0	>100B Bulgarian tokens; NRA integration; planned BRAIN++ data lake
Models	16	0.55	8.8	BgGPT 3.0 on Gemma 3 = L2 (anchor 11); raw 0.55 after portfolio-depth deductions
Infrastructure	17	0.55	9.4	BRAIN++ AI Factory €90M + Discoverer+ (domestic); no SecNumCloud equivalent
Hardware	17	0.25	4.3	No indigenous design; NVIDIA via TSMC
Manufacturing	12	0.05	0.6	No fabs, packaging, or refining
Software	8	0.45	3.6	Strong open-source culture; COMPL-AI toolkit

Human capital	12	0.65	7.8	INSAIT at CSRankings #14 Europe, #1 Eastern Europe
Regulatory	8	0.55	4.4	AI Act applies automatically; limited rule-making weight
Pillar total	**100**		**~45.9**	

Modifiers, in order: rule-stability ×0.95 (low-moderate exposure as an EU state deliberately in the EU procurement fold) on Hardware (4.3→4.1) and Models (8.8→8.4), total ~45.2; alliance ×1.00 (neutral EU position) — unchanged; operating-model **+1** (INSAIT coherent, resourced, explicitly open; untested through a change of government). **Final ~46** — just inside Partially autonomous (45–64). What holds Bulgaria back is structural: no chip industry, no manufacturing; that gap closes only through Europe-wide infrastructure.

3.2 France — ~70

Europe's clearest example of sovereign AI: Mistral (valued €11.7B after the ASML-led round), SecNumCloud with qualified providers including S3NS, SiPearl building a home-grown CPU, €109B announced at the 2025 summit, and an active hand in writing the EU's rules.

Pillar	Weight	Raw	Score	Note
Data	10	0.70	7.0	Strong public data; excellent French corpus
Models	16	0.85	13.6	Mistral Large/Medium/Le Chat; Europe's only top-10 global firm
Infrastructure	17	0.80	13.6	UAE-MGX campus + Mistral Essonne + SecNumCloud + Alice Recoque exascale
Hardware	17	0.50	8.5	SiPearl Rhea1 sampling; still TSMC/ARM underneath
Manufacturing	12	0.15	1.8	No leading edge; STMicro mature nodes; Soitec wafers
Software	8	0.55	4.4	Strong FOSS; orchestration US-dominated
Human capital	12	0.85	10.2	Polytechnique, ENS, Inria, CNRS; returnee pipeline at Mistral
Regulatory	8	0.90	7.2	Summit host; AI Act leadership; SecNumCloud as exported standard
Pillar total	**100**		**~66.3**	

Modifiers, in order: rule-stability ×0.95 (low exposure — deliberately multi-supplier, with domestic model capability, but US-supplied accelerators) on Hardware (8.5→8.1) and Models (13.6→12.9), total ~65.2; alliance ×1.12 (Tier-1 ally) on Hardware, Models, Infrastructure (8.1 + 12.9 + 13.6 = 34.6 → 38.7), total ~69.4; operating-model **+1** (Mistral as national champion — disciplined and state-aligned, untested through upheaval). **Final ~70**

— Strategically autonomous. Note what the two multipliers did: alliance added ~4 points of borrowed access, and rule-stability took ~1 back for the exposure that comes with it. What keeps France below 80 is manufacturing and home-grown accelerators. (Earlier revisions added an ad-hoc “+5 rule-setting bonus” here; v0.7.1 removes it — rule-making strength is already priced in the 0.90 Regulatory raw score.)

3.3 UAE — ~57

The most interesting case outside the US and China: cheap energy, deep state funds, and political flexibility, channelled through G42 into genuinely cutting-edge infrastructure.

Pillar	Weight	Raw	Score	Note
Data	10	0.50	5.0	Arabic via MBZUAI; weak open-data regime
Models	16	0.70	11.2	Falcon open-source series; Jais; +2 reasoning bonus included
Infrastructure	17	0.85	14.5	Stargate-UAE 1 GW (first 200 MW 2026); 5 GW campus; Khazna; energy strength (Barakah + solar) is inside this raw score
Hardware	17	0.45	7.7	~35,000 GB300 via G42; access is political, not owned
Manufacturing	12	0.15	1.8	No advanced node
Software	8	0.40	3.2	Azure and Oracle dependency
Human capital	12	0.60	7.2	MBZUAI operational; imported talent; small citizen base
Regulatory	8	0.75	6.0	US—UAE AI Acceleration Partnership; assurance agreement
Pillar total	**100**		**~56.6**	

Modifiers, in order: rule-stability **×0.85** (high exposure — access rests on case-by-case US permission, as the modifier table itself uses the UAE to illustrate) on Hardware (7.7→6.5) and Models (11.2→9.5), total ~53.8; alliance ×1.08 (special partnership) on Hardware, Models, Infrastructure (6.5 + 9.5 + 14.5 = 30.5 → 33.0), total ~56.2; operating-model **+1** (G42 as a coherent, well-funded state-linked vehicle with explicit sovereignty intent; the arrangement has weathered the China-pullback pivot but not a genuine access crisis). **Final ~57** — mid Partially autonomous. (Earlier revisions reached ~62 via an ad-hoc “+5 energy bonus” and by skipping rule-stability; v0.7.1 applies the framework’s own rules. Energy is already credited inside the 0.85 Infrastructure raw score, and the UAE’s defining exposure — access by permission — is exactly what the rule-stability modifier exists to price.) The honest reading: the UAE is building infrastructure faster than Europe, and remains one deal-by-deal decision away from losing the access that infrastructure depends on. The gaps are models at L2—L3, manufacturing, and a small citizen talent base.

Part IV. Case study: INSAIT and BgGPT

BgGPT is the clearest real-world example of **L2** on the Models ladder — how a small economy gains real control over its models without the cost of training from scratch.

4.1 BgGPT in brief

INSAIT (Sofia, part of Sofia University) builds BgGPT, a family of Bulgarian-language models: v0.1/0.2 on Mistral-7B (March 2024); BgGPT 1.0 on Gemma 2 (November 2024); BgGPT 3.0 on Gemma 3 in 4B/12B/27B sizes (March 2026) — multimodal, ~131k context, trained on >100B Bulgarian tokens plus balanced English. Its Branch-and-Merge method (EMNLP 2024) prevents catastrophic forgetting while adding a language. On Bulgarian tasks the 27B model holds its own against models three times its size and, for Bulgarian chat, against leading commercial models on INSAIT's benchmarks. Distributed in full precision, GPTQ W4A16, and GGUF (runnable locally via llama.cpp). First large institutional user: the National Revenue Agency, 7,000+ staff.

4.2 Why BgGPT is exactly L2

Level	Anchor	Why BgGPT is / is not at this level
L0	0	BgGPT runs on own machines with open weights — no outside provider
L1	4	More than a local copy: further trained on >100B Bulgarian tokens
L2	**11**	**Exactly this: heavy continued pre-training + instruction tuning on Gemma 3 weights, via Branch-and-Merge**
L3	13	Architecture and tokeniser are Gemma 3's — no structural modifications
L4	16	Starting weights come from Google; no from-scratch training

4.3 AI-SAF-N profile for INSAIT / BgGPT

Scored as an institutional profile (the institute, not the country — Bulgaria's national profile in Part III applies portfolio-depth deductions this focused profile does not):

Pillar	Score	Max	Rationale
Data	7	10	>100B curated Bulgarian tokens; instruction data
Models	11	16	L2 anchor (fine-tune/CPT on Gemma 3); Branch-and-Merge
Infrastructure	9	17	H100 compute via partnerships; limited own capacity
Hardware	4	17	Full NVIDIA/CUDA dependency
Manufacturing	1	12	No foundry access
Software	4	8	PyTorch + open inference; open-source contributions
Human capital	8	12	Top-venue publications; Sofia University pipeline
Regulatory	4	8	EU membership; rule-taker internationally
Total	**48**	**100**	Upper end of Partially autonomous

4.4 Effect on a Bulgarian enterprise

For a Bulgarian company, BgGPT opens a genuine new choice. Two options for a typical firm in a regulated industry:

Pillar	Scenario A: foreign API only	Scenario B: local BgGPT 27B
Data	3 (residency risk)	8 (data stays local)
Models	0 (L0 — foreign API)	11 (L2 — fine-tune capable)
Infrastructure	3 (cloud dependent)	6 (local inference feasible)
Software	1 (API wrapper)	4 (vLLM / llama.cpp expertise)
Other five pillars	~10	~10
Total	**~17**	**~39**

Moving from A to B adds about 22 points — from Fully dependent to Dependent-with-reserves, nearly Partially autonomous: the single biggest improvement available to a Bulgarian company without heavy spending on data centres or a large talent build.

4.5 Strategic implications

L2 is within reach for a small economy — continued pre-training plus instruction tuning matches commercial quality for tens of millions of euros, not hundreds. **People are the multiplier** — INSAIT's results come from a core team of roughly 20–30, which is why Human capital carries 12 points. **Resilience stays limited until the lower layers diversify** — BgGPT fixes the Models pillar; NVIDIA and CUDA dependence underneath remains, so the Version Deadlock risk shifts down a layer, to hardware export controls, where it unfolds more slowly. **The template transfers** — a state-backed academic centre + continued pre-training on the strongest open base + public weights is directly transposable to Portugal, Greece, Finland, the Baltics, Hungary, and most states with 5–15 million speakers, and raises an AI-SAF-N score by 5–10 points for a modest outlay.

For the master glossary, see Document 1.

Document 4 · State of Play

The dated market snapshot. Factual snapshot: August 2026. This document ages by design — it is refreshed each revision so the rubrics in Documents 2 and 3 do not have to be. Sources for every dated claim: Document 5, Evidence Base.

1. Models

Frontier and open weights. Capable open-weights models are cheap and common: Llama 4, Gemma 3, Qwen 3, DeepSeek V3/V4, Mistral. DeepSeek-V4 (April 2026 preview; reported 1.6T-parameter MoE, ~1M-token context, MIT-licensed weights) shipped with day-0 adaptation to Huawei Ascend, Cambricon, Hygon, and Moore Threads — the first top-tier model optimised for a fully non-NVIDIA stack rather than treating it as fallback.

Reasoning-capable models (DeepSeek R1, EXAONE Deep, Sarvam-M, Qwen thinking mode) are now a separate marker of national capability.

The Fable 5 / Mythos 5 episode (June—July 2026). A US export-control directive suspended all foreign-national access to Anthropic's frontier models on 12 June; the directive was lifted 30 June and Fable 5 restored globally 1 July, with Mythos 5 limited to approved US organisations. The 19-day outage is the framework's flagship live Version Deadlock case (Framework §3.4) and drove a wave of European sovereign-AI initiatives: Austria's proposal that the EU host Anthropic capacity, the UK's Isambard-trained "Lumen Sovereign" frontier model, and accelerated German investment around Aleph Alpha.

National models. BgGPT 3.0 (Bulgaria, March 2026) on Gemma 3; Meltemi (Greece), Sarvam-M (India), TAIDE (Taiwan) at L2; EuroLLM, Teuken, Poro, Aya-Expanse, EXAONE at L3; Falcon/Jais (UAE), Fugaku-LLM (Japan), DeepSeek/Qwen-on-Ascend (China) at L4. Mistral remains Europe's only top-10 global model firm (€11.7B valuation after the September 2025 ASML-led €1.7B Series C).

Reference costs. DeepSeek-V3's reported training cost (~$5.6M rented compute, 2.788M H800 GPU-hours) is the floor for replicating a top model; ~$500M/year approximates a full national ecosystem.

2. Cloud and sovereign infrastructure

Certified sovereign tier. SecNumCloud 3.2 qualified providers: OVHcloud, Outscale, Scaleway, and S3NS (Thales + Google, qualified 17 December 2025 — the first major US-technology service to pass); Bleu (Capgemini + Orange + Microsoft) still in qualification.

EU-operated US subsidiaries. AWS European Sovereign Cloud (GA 15 January 2026, Brandenburg, €7.8B): EU staff, EU-citizen directors, but Amazon-owned — CLOUD Act exposure arguably remains; qualifies at the BSI C5 tier. Azure EU Data Boundary is a contractual promise; Microsoft's own French Senate testimony (June 2025) conceded CLOUD Act protection cannot be guaranteed.

The EU policy layer. The Cloud and AI Development Act (proposed 3 June 2026, part of the Technological Sovereignty Package with Chips Act 2.0) would create a four-level Cloud Sovereignty Framework for public procurement and aims to triple EU data-centre capacity in 5—7 years; in trilogue. The Commission's first sovereignty-criteria cloud procurement (€180M, April 2026) went to four European groups. EUCS remains stalled with its sovereignty tier removed.

Compute programmes. 19 EuroHPC AI Factories across 16 states (~€2.6—2.7B committed) plus 13 Antennas. The AI Gigafactories call opened 30 July 2026: seven sites, ≥100,000 advanced chips each, €10B public targeting ≥€20B private; bids close 12 November 2026, awards early 2027, operations ~18 months after. Ten member states have signalled hosting interest. Outside the EU: Stargate UAE (1 GW, first 200 MW due Q3 2026) inside the 5 GW UAE—US AI Campus.

Constraints. Roughly 95% of commercial AI compute is US- or Chinese-owned. European data centres cluster in FLAP-D plus Madrid and Milan; grid capacity is a live constraint (Irish data centres exceeded 20% of national electricity in 2024).

3. Hardware

NVIDIA. Blackwell GB300 in full production; Vera Rubin platform launched at CES 2026 (NVL72 rack: 3.6 EFLOPS inference, ~$8.8M/rack, partner availability H2 2026). Vendor figures, not independently tested.

AMD. MI300X/MI325X in production at scale (Azure, Meta, El Capitan); MI355X clusters at Oracle up to 131,072 chips; ~$100B Meta—AMD deal (late 2025); MI400/Helios expected H2 2026. ROCm 7.x makes AMD a believable second source.

Hyperscaler ASICs — the real threat to NVIDIA's grip. Google TPU v7 Ironwood (GA November 2025; a reported 1M-chip Anthropic order); AWS Trainium3 (3nm; 500k-chip Anthropic cluster); Microsoft Maia 200; Meta MTIA v4 (RISC-V, with Broadcom). TrendForce forecasts custom-ASIC shipments growing 44.6% in 2026 vs 16.1% for GPUs, with NVIDIA's inference share possibly falling to 20—30% by 2028.

Chinese stack. Huawei Ascend 910C (~H100-class on paper; ~600k units 2026); Ascend 950PR shipping, 950DT due Q4 2026; Cambricon ~500k chips. With DeepSeek-V4's day-0 Ascend adaptation, China's top models no longer need CUDA.

European and Korean alternatives. SiPearl Rhea1 sampling (JUPITER); Rhea2 for Jules Verne; EU DARE RISC-V programme (~€240M); Tenstorrent (open RISC-V, Blackhole shipping); Rebellions ($400M at $2.3B pre-IPO); Cerebras ($10B+/750 MW OpenAI partnership, January 2026).

US export-control posture. Seven major turns 2022—2026, now deal-by-deal: the H20 15% and H200 25% revenue-share deals (2025); case-by-case licence review for H200-class (January 2026); the offshore-subsidiary rule (June 2026 — licensing requirements follow Chinese-headquartered firms abroad); H20 licences granted (July 2026). Uptake is another matter: as of mid-May 2026, no H200s had reportedly shipped to approved Chinese buyers.

4. Manufacturing and materials

Leading edge. TSMC N2 in mass production (Q4 2025; ~$25—27k/wafer; booked through 2028; Apple holds ~half of 2026—27 supply). Samsung 2nm ramping with yield struggles. Intel 18A ramping (Panther Lake shipped; AWS and Microsoft as customers) — but Intel cancelled Magdeburg and Wrocław (July 2025), forfeiting ~€10B in subsidies; ESMC Dresden (TSMC-backed, 28/22 + 16/12nm) proceeds. Chips Act 2.0 (proposed 3 June 2026) is the EU's response: faster permitting (≤12 months), investment conditions, "Grand Challenges" including AI chips. The European Court of Auditors projects the EU at ~11.7% of the global chip value chain by 2030, versus the 20% target.

Packaging and memory — the real bottlenecks. CoWoS: ~70k wafers/month end-2025, targeting 130k end-2026, NVIDIA holding ~half. HBM4: SK Hynix, Micron, Samsung all booked for 2026; the main constraint on Rubin-class parts.

Critical minerals. China refines ~89% of rare earths and ~60% of germanium. The October 2025 controls were paused in November 2025 for 12 months — but enforcement through 2026 has been active: new restrictions on 10 US companies (June) and 14 EU firms (July), formalised violation-handling rules, and the first detentions of foreign nationals over alleged rare-earth smuggling. The IEA warns full enforcement could put $6.5T of downstream production at risk. The EU is standing up a joint critical-minerals purchasing and stockpiling centre. The Nexperia affair (Dutch state supervision from September 2025; Chinese counter-controls; the company operationally split by March 2026) shows the same weapon pointed both ways.

5. Software

CUDA and alternatives. NVIDIA holds ~75—90% of the accelerator market; CUDA beats ROCm by 10—30% on compute-bound work, less on memory-bound. TorchTPU (Google + Meta, December 2025) brings full PyTorch support to TPUs — the most direct assault on CUDA lock-in. PyTorch's multi-party foundation governance (AMD, AWS, Google, Meta, Microsoft, NVIDIA on the board) makes single-vendor weaponisation harder.

Inference runtimes. vLLM: best TTFT, broad hardware support — the default. TensorRT-LLM: 10—30% more throughput on pure-NVIDIA deployments. SGLang: RadixAttention, up to ~6.4× throughput on shared-prefix/agent workloads. HuggingFace TGI: maintenance mode — avoid.

6. Regulatory timeline (as of August 2026)

Instrument	Status
EU AI Act — bans, AI literacy	In force since 2 February 2025
EU AI Act — GPAI obligations	In force 2 August 2025; **enforceable (AI Office penalty powers, Article 50 transparency) since 2 August 2026**
EU AI Act — high-risk	**Deferred by Digital Omnibus (in force 27 July 2026): Annex III → 2 December 2027; Annex I → 2 August 2028**
GPAI Code of Practice	Endorsed 1 August 2025; Meta declined; xAI signed safety chapter only
EU Data Act Chapter VII	Fully applicable since September 2025
DORA	In force since 17 January 2025
Council of Europe AI Convention (CETS 225)	In force 1 November 2025
CADA + Chips Act 2.0	Proposed 3 June 2026; in trilogue
New Delhi Declaration	Adopted 18—19 February 2026; 92 endorsements

US posture	Deal-by-deal export diplomacy; Fable/Mythos directive (12 June) lifted 30 June 2026

7. Human capital and talent reference figures

MacroPolo Global AI Talent Tracker 2.0 (March 2024, NeurIPS 2022 data): 75% of top AI researchers at US institutions did undergraduate work in the US or China; China produces 47% of top researchers by undergraduate origin; mobility is falling. Stanford AI Index 2026: the US holds ~75% of global GPU compute (up from 51%); China's share fell to ~14%; the US produced 50 notable models in 2025 vs China's 30; Switzerland (110.5) and Singapore (109.5) lead researchers per 100k population; the US—China top-model capability gap has all but closed; 80 of the 95 most notable 2025 models shipped without training code; average Foundation Model Transparency Index score fell to 40.

Document 5 · Evidence Base

Primary sources and verified citations. Base snapshot April 2026; entries marked ***[Added v0.7.1]*** *verified August 2026.*

How to use this document

This is the source list behind the AI-SAF Framework package. It gives the original source for every dated event, number, and named person mentioned in the three main framework documents (the Framework, the AI-SAF-N Field Guide, and the AI-SAF-C Field Guide) and in the companion refinement paper. It is built to do three jobs. First, it is a companion reference: if you come across an event in any framework document, you can find it here by topic and date and check the original source. Second, it stands on its own as a briefing, grouped by topic rather than by framework building block — so someone interested in, say, the home-grown cloud response to the CLOUD Act can read that section without knowing how the AI-SAF building blocks are organised. Third, it is the paper trail behind anything published: if a framework document is ever submitted to a journal, a policy forum, or a standards body, the sources can be pulled straight from here without checking them all over again. Every entry is laid out the same way. A short statement of the claim, in the same wording the framework documents use, is followed by one or more original sources with web links. A final line in italics shows which framework documents rely on that claim. Where a claim was checked against several sources, the most reliable one comes first. Where a claim could only be partly confirmed, a Note line is added.

Verification methodology

Every claim was checked by going straight to the original source (official press releases, court rulings, regulator decisions, laws, or peer-reviewed papers) or, where the original was behind a paywall or unavailable, by confirming it across at least two independent second-hand sources (major business press, legal commentary, or standards bodies). No claim rests on a single second-hand source. Three cautions about how this was put together.

First, the field moves faster than the sources can keep up: items confirmed here in April 2026 will eventually be out of date; the entries marked [Added v0.7.1] were verified in August 2026 and the same caution applies to them. Second, performance claims from vendors (FLOPS, memory bandwidth, benchmark scores) should be read as the maker's own specs, not as figures someone independently reproduced. Third, the scoring examples for specific firms (BNP Paribas, ING) in the framework documents are the authors' own estimates from public information — they show how the method works, and are not official outside assessments.

Document cross-reference key

In each entry, the "Referenced in" line uses the following abbreviations: Framework — AI-SAF Framework (Document 1) AI-SAF-C — AI-SAF-C Field Guide (Document 2, corporate scoring) AI-SAF-N — AI-SAF-N Field Guide (Document 3, national scoring) Refining — Refining AI-SAF v2.0 whitepaper (cited version)

Contents

Part A. Sovereignty incidents — cloud & data

Part B. Sovereignty incidents — hardware, export controls, supply chain

Part C. Sovereignty incidents — manufacturing & critical materials

Nexperia state supervision and supply standoff (September 2025 → ongoing) [Added v0.7.1]

Claim. In September 2025 the Netherlands invoked the Goods Availability Act to place Nexperia — the Nijmegen-headquartered semiconductor maker majority-owned by China's Wingtech — under temporary state supervision on national-security grounds. China responded with export controls on China-produced Nexperia chips, disrupting automotive production worldwide from October 2025. By March 2026 the Dutch and Chinese arms were operating as separate, often hostile entities (on 3 March 2026 Nexperia B.V. cut Chinese staff off from global IT systems). Partial de-escalation followed — China walked back export controls and The Hague relinquished oversight — but the entities had not reunited as of mid-2026. A textbook two-directional sovereignty incident: European assertion of control over a chip asset, answered by Chinese leverage over the downstream supply chain.

Sources: Z2Data, "The Nexperia Chip Crisis, Explained" https://www.z2data.com/insights/the-nexperia-chip-crisis-explained/; Jill Goldenziel, "The Dutch Seized a Chinese Chipmaker. Supply Chain Chaos Has Just Begun," Forbes, 31 December 2025 https://www.forbes.com/sites/jillgoldenziel/2025/12/31/the-dutch-seized-a-chinese-chipmaker-supply-chain-chaos-has-just-begun/; Diplomat Magazine, "China—Netherlands Technology Dispute over Nexperia Enters a Negotiated Phase," 6 January 2026 https://diplomatmagazine.eu/2026/01/06/china-netherlands-technology-dispute-over-nexperia-enters-a-negotiated-phase/.

Note. This incident predates the April 2026 snapshot but was absent from earlier Evidence Base revisions; added in v0.7.1.

Referenced in: AI-SAF-N Manufacturing pillar; State of Play §4.

China rare-earth enforcement wave (June — July 2026) [Added v0.7.1]

Claim. Despite the November 2025 pause, enforcement stayed active through 2026. On 22 June 2026 MOFCOM placed 10 US companies under new export restrictions; by 24 July it had blocked dual-use shipments to 14 EU firms. Announcement No. 26 (24 June 2026) formalised the handling of violations involving strategic-mineral dual-use controls. Two Japanese nationals were detained in Dalian (May 2026) over alleged rare-earth-related smuggling — among the first foreign-national detentions tied to these controls. The IEA warned that full enforcement could put $6.5 trillion of downstream production at risk.

Sources: Morgan Lewis, "Recent China Export Control Actions Signal Active Enforcement for Rare Earths and Strategic Minerals," July 2026 https://www.morganlewis.com/pubs/2026/07/recent-china-export-control-actions-signal-active-enforcement-for-rare-earths-and-strategic-minerals; China Briefing, "China's Rare Earth Export Controls — Impact on Businesses and Industries" https://www.china-briefing.com/news/chinas-rare-earth-export-controls-impacts-on-businesses/.

Referenced in: AI-SAF-N Manufacturing pillar (critical-minerals sub-indicator); State of Play §4.

Part D. Sovereign stack responses

EuroHPC AI Gigafactories call (30 July 2026) [Added v0.7.1]

Claim. The EuroHPC Joint Undertaking opened the formal call for seven AI Gigafactories on 30 July 2026: €10 billion in public funding intended to mobilise at least €20 billion more in private investment, each site to hold at least 100,000 advanced AI chips. Bids close 12 November 2026; awards are expected in early 2027, with operations targeted about 18 months after selection. Ten member states signalled hosting interest (Germany, Italy, France, Poland, Czechia, Denmark, Finland, Greece, Portugal, Spain).

Sources: EuroHPC JU, "The EuroHPC Joint Undertaking launches the AI Gigafactories Call," 30 July 2026 https://www.eurohpc-ju.europa.eu/eurohpc-joint-undertaking-launches-ai-gigafactories-call-2026-07-30_en; European Commission, "EU launches AI Gigafactories call" https://digital-strategy.ec.europa.eu/en/news/eu-launches-ai-gigafactories-call-boost-europes-computing-capacity-and-unlock-more-eu30-billion; Euronews, 30 July 2026 https://www.euronews.com/my-europe/2026/07/30/eu-opens-call-for-seven-gigafactories-to-train-next-generation-ai-technologies.

Referenced in: AI-SAF-N Infrastructure pillar (Local compute, pooled-capacity crediting); Framework §2.4 calibration; State of Play §2.

First EU sovereignty-criteria cloud procurement (April 2026) [Added v0.7.1]

Claim. The European Commission awarded a €180 million sovereign-cloud contract to four European provider groups (Post Telecom, StackIT, Scaleway, Proximus), reported as the first EU procurement run on explicit sovereignty criteria.

Sources: Julien Simon, "Two Sovereign Clouds, One Legal Wall," Medium https://julsimon.medium.com/two-sovereign-clouds-one-legal-wall-ff39bcc0432b.

Note. Secondary sourcing only; verify against the Commission's award notice before citing in anything published.

Referenced in: AI-SAF-N Infrastructure pillar; State of Play §2.

Part E. Regulatory instruments

EU AI Act — Digital Omnibus deferral and GPAI enforcement (27 July / 2 August 2026) [Added v0.7.1]

Claim. The Digital Omnibus on AI (proposed 19 November 2025) entered into force on 27 July 2026, deferring the AI Act's high-risk obligations: stand-alone Annex III systems to 2 December 2027, and AI embedded in Annex I regulated products to 2 August 2028. The GPAI provisions were not deferred: on 2 August 2026, Article 50 transparency obligations, Commission penalty powers over GPAI providers, and full national market-surveillance authority took effect as originally scheduled.

Sources: Gibson Dunn, "EU AI Act Omnibus Agreement — Postponed High-Risk Deadlines and Other Key Changes" https://www.gibsondunn.com/eu-ai-act-omnibus-agreement-postponed-high-risk-deadlines-and-other-key-changes/; Freshfields, "EU AI Act unpacked #34: The final Digital Omnibus on AI" https://www.freshfields.com/en/our-thinking/blogs/technology-quotient/eu-ai-act-unpacked-34-the-final-digital-omnibus-on-ai-key-amendments-to-the-a-102nber; DLA Piper, "The Digital AI Omnibus: Proposed deferral of high-risk AI obligations" https://knowledge.dlapiper.com/dlapiperknowledge/globalemploymentlatestdevelopments/2026/The-Digital-AI-Omnibus-Proposed-deferral-of-high-risk-AI-obligations-under-the-AI-Act.

Referenced in: Framework Part V; AI-SAF-C Regulatory pillar; AI-SAF-N Regulatory pillar; State of Play §6.

EU Technological Sovereignty Package: CADA and Chips Act 2.0 (3 June 2026) [Added v0.7.1]

Claim. On 3 June 2026 the European Commission adopted the Cloud and AI Development Act proposal (COM(2026) 502) and the Chips Act 2.0 proposal (COM(2026) 504). CADA aims to triple EU data-centre capacity within five to seven years, creates data-centre acceleration zones, and establishes a Cloud Sovereignty Framework grading cloud services on four assurance levels — from EU data location (level 1) to full supply-chain control free of third-country interference (level 4) — applicable to public-sector procurement. Chips

Act 2.0 targets permitting within 12 months, investment conditions, and "Grand Challenges" including AI chips; the European Court of Auditors projects the EU at ~11.7% of the global chip value chain by 2030 versus the 20% target. Both proposals are in trilogue.

Sources: European Commission, "Proposal for the Cloud and AI Development Act (CADA)" https://digital-strategy.ec.europa.eu/en/library/proposal-cloud-and-ai-development-act-cada; EUR-Lex COM(2026) 502 https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=COM%3A2026%3A502%3AFIN; Covington, "The EU Cloud and AI Development Act in Depth," 11 June 2026 https://www.insideglobaltech.com/2026/06/11/the-eu-cloud-and-ai-development-act-in-depth/; European Commission, "Proposal for the Chips Act 2.0" https://digital-strategy.ec.europa.eu/en/library/proposal-chips-act-20.

Referenced in: Framework Part V; AI-SAF-C Infrastructure pillar; AI-SAF-N Infrastructure and Manufacturing pillars; State of Play §2, §4.

New Delhi AI Impact Summit and Declaration (16—21 February 2026) [Added v0.7.1 — promoted from Part J]

Claim. The India AI Impact Summit ran 16—21 February 2026 at Bharat Mandapam, New Delhi, with representatives of 118 countries. The New Delhi Declaration on AI Impact, adopted 18—19 February, was endorsed by 92 countries and international organisations, structured around seven pillars ("Chakras"). Outcomes include the Charter for the Democratic Diffusion of AI, the Global AI Impact Commons, and India's GPU expansion (20,000 added to an existing 38,000, targeting 100,000 by end-2026).

Sources: Press Information Bureau (Government of India), "AI Impact Summit 2026 Concludes with Adoption of New Delhi Declaration" https://www.pib.gov.in/PressReleasePage.aspx?PRID=2231208&v=4&reg=3&lang=2; Ministry of External Affairs press release https://www.mea.gov.in/press-releases.htm?dtl%2F40810%2FAI_Impact_Summit_2026_Concludes_with_Adoption_of_New_Delhi_Declaration=.

Referenced in: AI-SAF-N Regulatory pillar; State of Play §6.

Part F. Copyright and legal rulings

Part G. Research and technical references

Part H. Benchmark frameworks (peer indices)

Part J. Claims not yet individually verified

Part A. Sovereignty incidents — cloud & data

Microsoft France CLOUD Act testimony (10 June 2025)

Claim. Anton Carniaux, Director of Public and Legal Affairs at Microsoft France, testified under oath to a French Senate inquiry into public buying and digital sovereignty that he could not guarantee French citizens' data held in Microsoft's EU data centres would never be handed to US authorities without France's consent. Asked directly, he said "No, I cannot guarantee that." The law that makes this possible is the US CLOUD Act of 2018.

Sources: Sénat de la République française, Commission d'enquête sur la commande publique et la souveraineté numérique, Audition d'Anton Carniaux, 10 June 2025; Lindsay Clark, "Microsoft exec admits it cannot guarantee data sovereignty," The Register, 25 July 2025 https://www.theregister.com/2025/07/25/microsoft_admits_it_cannot_guarantee; Cristiano Lima-Strong, "Microsoft tells French lawmakers it can't protect user data from US demands," SDxCentral, 21 July 2025 https://www.sdxcentral.com/news/microsoft-tells-french-lawmakers-it-cant-protect-user-data-from-us-demands/.

Referenced in: Framework §2 (illustrative incident), AI-SAF-N Infrastructure pillar, AI-SAF-C Infrastructure pillar, Refining §3.3

Microsoft / ICC email block of Karim Khan (February—May 2025)

Claim. After the US Treasury imposed sanctions in February 2025 on International Criminal Court Chief Prosecutor Karim Khan — in response to ICC investigations into Israeli officials — Khan's Microsoft email was switched off. The ICC then moved its office and teamwork software to OpenDesk, a free, open-source platform from Germany's Centre for Digital Sovereignty (ZenDiS). Microsoft has disputed some accounts of how events unfolded.

Sources: Associated Press reporting, February—May 2025; Computer Weekly, "Microsoft's ICC email block reignites European data sovereignty concerns," 23 May 2025 https://www.computerweekly.com/opinion/Microsofts-ICC-email-block-reignites-European-data-sovereignty-concerns; Xinhua, "Microsoft email block of ICC prosecutor fuels Dutch alarm over US tech dependence," 20 May 2025 https://english.news.cn/20250521/4a278fdce8324af59346ecf47ac407c9/c.html; Lindsay Clark, "Microsoft throws spox under the bus in ICC email flap," The Register, 18 February 2026 https://www.theregister.com/2026/02/18/microsoft_asks_uk_parliament_to_correct_record/.

Referenced in: Framework §2, AI-SAF-C Infrastructure, Refining §3.3

Meta Llama multimodal EU withdrawal (18 July 2024)

Claim. Meta said it would not release its upcoming multimodal Llama model (one that handles text, images, and more) in the European Union, blaming "the unpredictable nature of the European regulatory environment." Its specific complaint was about GDPR duties around using EU users' data to train AI, and it came after the Irish Data Protection Commission asked Meta to hold off on plans to use public Facebook and Instagram content for training.

Sources: Ryan Heath, "Scoop: Meta won't offer future multimodal AI models in EU," Axios, 17 July 2024 https://www.axios.com/2024/07/17/meta-future-multimodal-ai-models-eu; Suhasini Srinivasaragavan, "Meta will not launch multimodal Llama AI model in EU," Silicon Republic, 18 July 2024 https://www.siliconrepublic.com/business/meta-llama-3-multimodal-ai-model-llm-eu-europe-launch-delay.

Referenced in: Framework §2, AI-SAF-N Data pillar, AI-SAF-N Regulatory pillar, Refining §3.1

Apple Intelligence EU delay (October 2024 → April 2025)

Claim. Apple Intelligence features arrived worldwide with iOS 18.1 in October 2024 but were held back for EU users until iOS 18.4 in April 2025, expressly because of the Digital Markets Act's rules requiring products to work with rivals' systems.

Sources: Romain Dillet, "Apple Intelligence is coming to the EU in April 2025," TechCrunch, 28 October 2024 https://techcrunch.com/2024/10/28/apple-intelligence-is-coming-to-the-eu-in-april-2025/; Apple Inc., "The Digital Markets Act's impacts on EU users," Apple Newsroom, September 2025 https://www.apple.com/newsroom/2025/09/the-digital-markets-acts-impacts-on-eu-users/.

Referenced in: Framework §2, AI-SAF-N Data pillar, Refining §3.1

Italian Garante DeepSeek ban (30 January 2025)

Claim. Italy's data protection authority (the Garante) ordered DeepSeek — Hangzhou DeepSeek AI and Beijing DeepSeek AI — to stop handling Italian users' personal data after the companies claimed they "did not operate in Italy" and that European rules did not apply. The Garante called the companies' response "completely insufficient." The order covered the app-store versions; the website stayed reachable. Authorities in Belgium, Ireland, France, Portugal, and Spain opened their own investigations.

Sources: Garante per la protezione dei dati personali, press release of 30 January 2025; Bird & Bird, "The Garante imposes a definitive limitation on the processing of Italian users' personal data," 26 February 2025 https://www.twobirds.com/en/insights/2025/the-garante-imposes-a-definitive-limitation-on-the-processing-of-italian-users'-personal-data; Euronews, "DeepSeek AI blocked by Italian authorities," 31 January 2025 https://www.euronews.com/next/2025/01/31/deepseek-ai-blocked-by-italian-authorities-as-others-member-states-open-probes.

Referenced in: Framework §2, AI-SAF-N Regulatory pillar, Refining §3.8

Part B. Sovereignty incidents — hardware, export controls, supply chain

Biden AI Diffusion Rule / Trump rescission (15 January → 13 May 2025)

Claim. The Biden administration issued the “AI Diffusion Rule” on 15 January 2025 (published in the Federal Register), set to take effect on 15 May 2025. The rule sorted the world into three tiers and capped each country’s imports of advanced chips. On 13 May 2025, two days before it would have kicked in, the Trump administration’s Commerce Department Bureau of Industry and Security (BIS) scrapped it, with Under Secretary Jeffrey Kessler telling BIS enforcement staff not to apply it. At the same time, BIS put out three new guidance documents — on controlling advanced-computing chips, on training AI for weapons-of-mass-destruction uses, and on tactics used to dodge supply-chain controls.

Sources: US Department of Commerce, BIS, “Department of Commerce Rescinds Biden-Era Artificial Intelligence Diffusion Rule,” 13 May 2025 https://media.bis.gov/press-release/department-commerce-rescinds-biden-era-artificial-intelligence-diffusion-rule-strengthens-chip-related; Becca Szkutak, “Trump administration officially rescinds Biden’s AI diffusion rules,” TechCrunch, 13 May 2025 https://techcrunch.com/2025/05/13/trump-administration-officially-rescinds-bidens-ai-diffusion-rules/.

Referenced in: Framework §2, AI-SAF-N Hardware pillar, AI-SAF-N Regulatory pillar, Refining §3.4

Nvidia H20 / AMD MI308 revenue-share deal (11 August 2025)

Claim. President Trump announced that Nvidia would pay the US government 15% of what it earns from selling H20 chips to China, with AMD agreeing to the same 15% on MI308 sales. This was a fundamental shift — from banning these exports to taxing them. In the previous quarter (April 2025), export controls had blocked roughly $2.5 billion of H20 sales to China from showing up in Nvidia’s results.

Sources: Fortune / Bloomberg, “Nvidia, AMD agree to pay US 15% of China chip sale revenue,” 10 August 2025 https://fortune.com/2025/08/10/nvidia-amd-chips-h20-mi308-china-sales-revenue-trump-export-license/; NPR, “Trump says Nvidia will hand the US 15% of its H20 chip sales to China,” 11 August 2025 https://www.npr.org/2025/08/11/nx-s1-5498689/trump-nvidia-h20-chip-sales-china; Bruno Micheli, “Trump’s Political Tax on Nvidia Chips to China,” American Action Forum, 12 August 2025 https://www.americanactionforum.org/insight/trumps-political-tax-on-nvidia-chips-to-china/.

Referenced in: Framework §2, AI-SAF-N Hardware pillar, Refining §3.4

Nvidia H200 revenue-share deal (8 December 2025)

Claim. President Trump announced that Nvidia would be allowed to ship H200 chips to approved customers in China, on condition that the US take a 25% cut of the revenue. Chinese President Xi Jinping reportedly responded favourably.

Sources: Arjun Kharpal, "Trump greenlights Nvidia H200 AI chip sales to China if US gets 25% cut," CNBC, 8 December 2025 https://www.cnbc.com/2025/12/08/trump-nvidia-h200-sales-china.html.

Referenced in: Framework §2, AI-SAF-N Hardware pillar, Refining §3.4

CrowdStrike global outage (19 July 2024)

Claim. A broken update to CrowdStrike's Falcon Sensor security software (channel file 291, timestamped 2024-07-19 04:09 UTC) set off what is widely called the largest IT outage in history. About 8.5 million Windows devices — roughly 1% of all Windows machines — crashed, throwing airlines, banks, hospitals, shops, and emergency services into chaos worldwide. Insurers estimated the cost to U.S. Fortune 500 companies at $5.4 billion. Delta Air Lines reported losses of about $500 million and sued CrowdStrike on 25 October 2024. In May 2025, a Georgia judge let Delta press claims of gross negligence, computer trespass, and limited fraud.

Sources: "2024 CrowdStrike-related IT outages," Wikipedia (consolidating contemporaneous press coverage) https://en.wikipedia.org/wiki/2024_CrowdStrike-related_IT_outages; Microsoft Corporation, "Helping our customers through the CrowdStrike outage," 20 July 2024 https://blogs.microsoft.com/blog/2024/07/20/helping-our-customers-through-the-crowdstrike-outage/; Sean Michael Kerner, "CrowdStrike outage explained: What caused it and what's next," TechTarget, updated September 2024 https://www.techtarget.com/whatis/feature/Explaining-the-largest-IT-outage-in-history-and-whats-next.

Referenced in: Framework §2, AI-SAF-N Software pillar, AI-SAF-C Software pillar, Refining §3.6

NVIDIA Vera Rubin platform (CES 2026, 5 January 2026)

Claim. NVIDIA formally launched the Vera Rubin platform at CES 2026. Rubin GPU: 336 billion transistors (two-reticle design), 50 PFLOPS NVFP4 inference, 35 PFLOPS NVFP4 training (vendor figures quoted as 5× and 3.5× Blackwell), 288 GB HBM4, 22 TB/s memory bandwidth. Vera CPU: 88 custom Olympus Arm v9.2 cores with Spatial Multi-Threading (176 threads), 1.5 TB LPDDR5x, NVLink-C2C at 1.8 TB/s. Vera Rubin NVL72 rack: 72 Rubin GPUs and 36 Vera CPUs delivering 3.6 EFLOPS NVFP4 inference and 2.5 EFLOPS NVFP4 training, 20.7 TB HBM4 capacity, 1.6 PB/s HBM bandwidth, 260 TB/s NVLink 6 rack-scale bandwidth. Full production reported Q1 2026; partner availability H2 2026. First cloud deployments announced at AWS, Google Cloud, Microsoft (Fairwater AI superfactories), OCI, CoreWeave, Lambda, Nebius, Nscale. Per-rack price reported in trade press at up to $8.8 million.

Sources: NVIDIA, “NVIDIA Kicks Off the Next Generation of AI With Rubin,” press release, 5 January 2026 https://nvidianews.nvidia.com/news/rubin-platform-ai-supercomputer; NVIDIA Technical Blog, “Inside the NVIDIA Vera Rubin Platform,” 5 January 2026 https://developer.nvidia.com/blog/inside-the-nvidia-rubin-platform-six-new-chips-one-ai-supercomputer/; Tom’s Hardware, “Nvidia launches Vera Rubin NVL72 AI supercomputer at CES,” 5 January 2026 https://www.tomshardware.com/pc-components/gpus/nvidia-launches-vera-rubin-nvl72-ai-supercomputer-at-ces-promises-up-to-5x-greater-inference-performance-and-10x-lower-cost-per-token-than-blackwell-coming-2h-2026.

Note. The per-rack price and the speed (FLOPS) figures come from NVIDIA itself; independent testing will follow in the second half of 2026.

Referenced in: Framework §2, AI-SAF-N Hardware pillar, Refining §3.4

Anthropic Fable 5 / Mythos 5 access suspension (12 June 2026)

Claim. On 12 June 2026 the US government, citing national-security export-control authority, issued a directive ordering Anthropic to suspend all access to its frontier Fable 5 and Mythos 5 models by any foreign national — whether inside or outside the United States, and including Anthropic’s own foreign-national employees. Anthropic disabled both models for all customers to comply, three days after Fable 5’s public release. Reported triggers were concern that the Mythos-class model could surface restricted cyber-attack and software-vulnerability information (raised after Amazon researchers elicited such output, escalated by CEO Andy Jassy) and suspected access by a China-linked group. The suspension cut off enterprises and governments worldwide overnight and reignited “sovereign AI” calls in Europe and India.

Sources: Anthropic, “Statement on the US government directive to suspend access to Fable 5 and Mythos 5,” 12 June 2026 https://www.anthropic.com/news/fable-mythos-access; Jordan Novet, “Anthropic disables access to Fable 5 and Mythos 5 to comply with government directive,” CNBC, 12 June 2026 https://www.cnbc.com/2026/06/12/anthropic-disables-access-to-fable-5-and-mythos-5-to-comply-with-government-directive.html; “US orders Anthropic to disable AI models for all foreign nationals,” Al Jazeera, 13 June 2026 https://www.aljazeera.com/news/2026/6/13/us-orders-anthropic-to-disable-ai-models-for-all-foreign-nationals; “How a warning from Amazon led the White House to shut down Anthropic’s Mythos model,” Fortune, 14 June 2026 https://fortune.com/2026/06/14/how-a-warning-from-amazon-led-the-white-house-to-shut-down-anthropics-mythos-model/; “US clampdown on Anthropic models sends EU sovereignty surge into overdrive,” The Register, 15 June 2026 https://www.theregister.com/ai-and-ml/2026/06/15/us-clampdown-on-anthropic-models-sends-eu-sovereignty-surge-into-overdrive/.

Note. Compiled from the originating report (The Verge, https://www.theverge.com/ai-artificial-intelligence/949986/anthropic-fable-mythos-shutdown-sovereign-ai) and cross-checked against the company statement and major business press above; wording on the directive's exact scope — especially its extraterritorial reach to foreign-national employees — should be re-verified as reporting settles.

Referenced in: Framework §1.1 (resilience), §2.3 (Models weight), Part III (Version Deadlock — live AI case), Part V (uncertainty), Conclusion; AI-SAF-N Models pillar and rule-stability modifier; AI-SAF-C Models pillar.

Fable 5 / Mythos 5 restoration (30 June — 1 July 2026) [Added v0.7.1]

Claim. On 30 June 2026 the US Department of Commerce lifted the 12 June export-control directive. Fable 5 was restored to all customers worldwide on 1 July, ending a 19-day suspension; Mythos 5 was reintroduced only to approved US organisations following government review. Anthropic stated that the jailbreak cited by the government was narrow and that comparable capabilities could be elicited from other publicly available models not subject to similar controls. Reported aftermath includes closer collaboration between Anthropic and US agencies on model testing, threat sharing, and jailbreak-risk standards, and continued policy debate over whether model-access export controls become a repeated instrument.

Sources: Anthropic / CNBC, "Anthropic says Trump admin has lifted export controls on Claude Fable 5 and Mythos 5," 30 June 2026 https://www.cnbc.com/2026/06/30/anthropic-says-trump-admin-has-lifted-export-controls-on-claude-fable-5-and-mythos-5.html; Al Jazeera, "US lifts restrictions on powerful AI models Fable and Mythos," 1 July 2026 https://www.aljazeera.com/economy/2026/7/1/us-lifts-restrictions-on-powerful-ai-models-fable-mythos-anthropic-says; TechPolicy.Press, "Did the US Government Just Set An AI Export Precedent by Blocking Mythos?" https://www.techpolicy.press/did-the-us-government-just-set-an-ai-export-precedent-by-blocking-mythos/; Fortune, "Anthropic shutdown ignites calls for sovereign AI across Europe," 16 June 2026 https://fortune.com/2026/06/16/anthropic-shutdown-sparks-global-scramble-for-sovereign-ai/ (including Austria's 28 June letter urging EU hosting of Anthropic capacity).

Referenced in: Framework §1.1, §2.3, §3.4 (restoration is not recovery), Part V; AI-SAF-C Models pillar (jurisdiction diversification); AI-SAF-N rule-stability modifier.

US chip-export posture codification (January — July 2026) [Added v0.7.1]

Claim. On 13 January 2026 the Commerce Department revised the licence-review posture for Nvidia H200- and AMD MI325X-class chips from presumption of denial to case-by-case review. In June 2026 BIS affirmed that its licensing requirements for advanced AI chips reach subsidiaries of Chinese-headquartered companies located outside China. In July 2026 Commerce said it would grant Nvidia export licences for the H20. Reported uptake has lagged policy: as of mid-May 2026, no H200 chips had shipped to the Chinese companies approved to buy them.

Sources: Mayer Brown, "Administration Policies on Advanced AI Chips Codified," January 2026 https://www.mayerbrown.com/en/insights/publications/2026/01/administration-policies-on-advanced-ai-chips-codified; Al Jazeera, "US says ban on AI chip shipments applies to Chinese firms outside China," 1 June 2026 https://www.aljazeera.com/economy/2026/6/1/us-says-ban-on-ai-chip-shipments-applies-to-chinese-firms-outside-china; Brookings, "Ball game's over — the US is out of the AI chip market in China" https://www.brookings.edu/articles/ball-games-over-the-us-is-out-of-the-ai-chip-market-in-china/.

Referenced in: Framework §3.1 (Hardware row), Part V; AI-SAF-N Hardware pillar and rule-stability modifier; State of Play §3.

DeepSeek-V4 with day-0 Huawei Ascend adaptation (April 2026) [Added v0.7.1]

Claim. DeepSeek released V4 in preview on 24 April 2026 with open weights under the MIT licence (reported 1.6-trillion-parameter Mixture-of-Experts design with ~1M-token context; figures pending verification against the technical report). Huawei Ascend (950PR/950DT), Cambricon, Hygon, and Moore Threads completed day-0 adaptation at launch — an architectural decision to treat the domestic Chinese stack as a first-class target rather than a fallback, and the strongest evidence to date that a top-tier Chinese model no longer requires NVIDIA's CUDA.

Sources: Pandaily, "DeepSeek V4 Completes Full Adaptation to Huawei Ascend," May 2026 https://pandaily.com/deepseek-v4-huawei-ascend-adaptation-may2026; Huawei Central, "DeepSeek V4 and Huawei Ascend: a day 0 partnership" https://www.huaweicentral.com/deepseek-v4-and-huawei-ascend-ai/; "DeepSeek (chatbot)," Wikipedia (consolidating coverage) https://en.wikipedia.org/wiki/DeepSeek_(chatbot).

Note. Parameter count, context length, and pricing circulate in trade press with some variance; verify against the DeepSeek technical report before relying on specific figures.

Referenced in: Framework Part V (uncertainty resolved); AI-SAF-N Models pillar (L4 example), Software pillar; State of Play §1, §3.

Part C. Sovereignty incidents — manufacturing & critical materials

Intel Magdeburg fab cancellation (24 July 2025)

Claim. Intel's new CEO, Lip-Bu Tan, formally cancelled Intel's planned €30+ billion "Silicon Junction" chip factory in Magdeburg, Germany (first announced in March 2022), along with a €4.6 billion assembly-and-test plant near Wrocław, Poland — part of a 15% staff cut down to about 75,000 employees. Intel said it had "decided not to move forward with previously planned projects in Germany and Poland" because "the company invested too much, too soon — without adequate demand." Intel gave up roughly €10 billion in German EU Chips Act subsidies and $1.9 billion in EU state aid earmarked for Poland.

Sources: Intel Corporation, Q2 2025 earnings announcement and CEO memo of 24 July 2025; Brussels Signal, "Intel cancels multi-billion euro chip factory in Germany," 25 July 2025 https://brusselssignal.eu/2025/07/intel-cancels-multi-billion-euro-chip-factory-in-germany/; heise online, “Intel gives up Magdeburg fab and announces end of foundry,” 25 July 2025 https://www.heise.de/en/news/Intel-gives-up-Magdeburg-fab-and-announces-end-of-foundry-10499170.html; EE Times Europe, “Intel’s Retreat Shifts Europe Semiconductor Reality,” 7 November 2025 https://www.eetimes.com/intel-retreat-shifts-europe-semiconductor-reality/.

Referenced in: Framework §2, AI-SAF-N Manufacturing pillar, Refining §3.5

China rare-earth export controls (9 October → 9 November 2025 pause)

Claim. China’s Commerce Ministry (MOFCOM) announcements Nos. 55—58, 61 and 62 (9 October 2025) sharply widened China’s controls on exporting rare-earth metals. After the Trump—Xi meeting in Busan, MOFCOM announcements Nos. 70 and 72 (7—9 November 2025) put those controls on hold until 10 November 2026. China refines roughly 89% of the world’s rare earths and controls almost the entire supply chain for high-performance rare-earth magnets. The earlier licensing rules of 4 April 2025, covering seven heavy rare-earth elements (scandium, yttrium, samarium, gadolinium, terbium, dysprosium, lutetium), stay in force. In November 2025, EU Commissioner Stéphane Séjourné announced plans for an EU critical-minerals centre to buy and stockpile these materials jointly.

Sources: Dorcas Wong and Qian Zhou, “China’s Rare Earth Export Controls — Impact on Businesses and Industries,” China Briefing, 12 November 2025 https://www.china-briefing.com/news/chinas-rare-earth-export-controls-impacts-on-businesses/; Clark Hill PLC, “China Hits Pause on Rare-Earth Export Controls” https://www.clarkhill.com/news-events/news/china-hits-pause-on-rare-earth-export-controls-and-what-it-means-for-supply-chains/; European Parliament Research Service, “China’s rare-earth export restrictions,” November 2025 https://www.europarl.europa.eu/thinktank/en/document/EPRS_ATA(2025)779220.

Referenced in: Framework §2, AI-SAF-N Manufacturing pillar, Refining §3.5

Part D. Sovereign stack responses

S3NS SecNumCloud 3.2 qualification (17—19 December 2025)

Claim. S3NS, a joint venture between Thales and Google Cloud, won SecNumCloud 3.2 certification from France’s cybersecurity agency ANSSI on 17 December 2025 for its PREMI3NS service. The service is run entirely by S3NS staff in France, with a “quarantine zone” that screens Google’s updates — making it the first major cloud service built on U.S. technology to earn SecNumCloud certification.

Sources: S3NS SAS, “S3NS Announces SecNumCloud Qualification for PREMI3NS,” press release, 19 December 2025 https://www.businesswire.com/news/home/20251218817208/en/S3NS-Announces-SecNumCloud-Qualification-for-PREMI3NS-its-Trusted-Cloud-Offering; Thales Group, “S3NS receives SecNumCloud qualification,” 19 December 2025 https://www.thalesgroup.com/en/news-centre/insights/group/s3ns-receives-secnumcloud-qualification-turning-point-trusted-cloud.

Referenced in: Framework §2, AI-SAF-N Infrastructure pillar, AI-SAF-C Infrastructure pillar, Refining §3.3

AWS European Sovereign Cloud general availability (15 January 2026)

Claim. The AWS European Sovereign Cloud became generally available on 15 January 2026, with its first data-centre region in Brandenburg, Germany, backed by a €7.8 billion commitment. It is run through EU-only German subsidiaries with EU-citizen managing directors (Stéphane Israël and Stefan Hoechbauer) and an EU-only advisory board. But the legal entity is still wholly owned by Amazon.com Inc., and commentators have pointed out that CLOUD Act exposure arguably remains.

Sources: Amazon Web Services, “AWS Launches AWS European Sovereign Cloud and Announces Expansion Across Europe,” press release, Potsdam, 15 January 2026 https://press.aboutamazon.com/aws/2026/1/aws-launches-aws-european-sovereign-cloud-and-announces-expansion-across-europe; AWS Blog, “Opening the AWS European Sovereign Cloud,” 14 January 2026 https://aws.amazon.com/blogs/aws/opening-the-aws-european-sovereign-cloud/; InfoQ, “AWS Launches European Sovereign Cloud amid Questions about US Legal Jurisdiction,” 22 January 2026 https://www.infoq.com/news/2026/01/aws-european-sovereign-cloud/.

Referenced in: Framework §2, AI-SAF-N Infrastructure pillar, AI-SAF-C Infrastructure pillar, Refining §3.3

EuroHPC AI Factories (December 2024 → October 2025)

Claim. The EuroHPC Joint Undertaking (a joint EU supercomputing program) has chosen 19 AI Factories across 16 EU member states, in three rounds: seven in December 2024 (Finland, Germany, Greece, Italy, Luxembourg, Spain, Sweden), six in March 2025 (Austria, Bulgaria, France, Germany, Poland, Slovenia), and six in October 2025 (Czech Republic, Lithuania, Netherlands, Romania, Spain, Poland). On top of that, 13 smaller AI Factory “Antennas” were chosen in Belgium, Cyprus, Hungary, Ireland, Latvia, Malta, Slovakia, Iceland, Moldova, North Macedonia, Serbia, Switzerland and the UK. The combined EU-and-member-state commitment is more than €2.6—2.7 billion.

Sources: EuroHPC JU, “AI Factories,” programme page https://www.eurohpc-ju.europa.eu/ai-factories_en; EuroHPC JU, “Six additional AI Factories to Expand Europe’s AI Capabilities,” 10 October 2025 https://www.eurohpc-ju.europa.eu/eurohpc-ju-selects-six-additional-ai-factories-expand-europes-ai-capabilities-2025-10-10_en; European Commission, “EU expands network of AI Factories” https://digital-strategy.ec.europa.eu/en/news/eu-expands-network-ai-factories-strengthening-its-ai-continent-ambition.

Referenced in: Framework §2, AI-SAF-N Infrastructure pillar, Refining §3.3

Stargate UAE (announced 22 May 2025; first phase Q3 2026)

Claim. Stargate UAE is a 1-gigawatt cluster of AI computing power being built by G42 (run by OpenAI and Oracle, with NVIDIA supplying Grace Blackwell GB300 systems, Cisco providing the networking, and SoftBank taking part) inside a planned 5-gigawatt UAE—US AI Campus. The first 200-megawatt phase is due to be finished in Q3 2026, according to Khazna Data Centres (December 2025). Reported investment is close to $20 billion. It was announced with UAE President Sheikh Mohamed bin Zayed Al Nahyan and U.S. President Donald Trump present.

Sources: G42 Inc., "Global Tech Alliance Launches Stargate UAE," press release, 22 May 2025 https://www.g42.ai/resources/news/global-tech-alliance-launches-stargate-uae; OpenAI, "Introducing Stargate UAE" https://openai.com/index/introducing-stargate-uae/; The National, "Stargate UAE's first phase to be completed in third quarter of 2026," 5 December 2025 https://www.thenationalnews.com/business/2025/12/05/stargate-uaes-first-phase-to-be-completed-in-third-quarter-of-2026/.

Referenced in: Framework §2, AI-SAF-N Infrastructure pillar, AI-SAF-N Hardware pillar, Refining §3.3

Mistral AI Series C (9 September 2025)

Claim. Mistral AI raised €1.7 billion in a Series C funding round led by ASML (which put in €1.3 billion for about an 11% stake once all shares are counted), valuing the company at €11.7 billion afterward. DST Global, Andreessen Horowitz, Bpifrance, Lightspeed, Nvidia and others also took part. It is the largest single private investment in a European AI model company so far, and puts France at Level 4 on the AI-SAF-N Models scale.

Sources: Mistral AI, "Mistral AI raises €1.7B to accelerate technological progress with AI," 9 September 2025 https://mistral.ai/news/mistral-ai-raises-1-7-b-to-accelerate-technological-progress-with-ai; Arjun Kharpal, "AI firm Mistral valued at $14 billion as chip giant ASML takes major stake," CNBC, 9 September 2025 https://www.cnbc.com/2025/09/09/ai-firm-mistral-valued-at-14-billion-as-asml-takes-major-stake.html.

Referenced in: Framework §2, AI-SAF-N Models pillar (France worked example), Refining §5.2

INSAIT BgGPT 3.0 (March 2026)

Claim. INSAIT released BgGPT 3.0 in three sizes — 4 billion, 12 billion, and 27 billion parameters — built on Google's Gemma 3 and further trained on more than 100 billion words of Bulgarian text. It can handle images as well as text, remembers up to about 131,000 tokens of context at once, was pre-trained on data through May 2025, and had its instruction tuning finished through October 2025. It comes in full precision, a compressed (GPTQ W4A16) version, and the GGUF format. The earlier BgGPT (2023) was based on Gemma 2 and used about 85 billion Bulgarian and 15 billion English words, trained with the "Branch-and-Merge" method that INSAIT researchers published at the EMNLP 2024 conference. INSAIT runs its own dedicated systems for projects with Bulgaria's National Revenue Agency and National Audit Office.

Sources: INSAIT, "INSAIT unveils a new generation of AI," press release, March 2026 https://insait.ai/insait-unveils-a-new-generation-of-ai-precision-models-for-business-and-an-upgraded-bggpt-chat-for-all-citizens/; Bulgarian News Agency (BTA), "Bulgaria Among Few Countries with Publicly Built Sovereign AI Offered Free to Citizens," March 2026; INSAIT-Institute/BgGPT-Gemma-3-27B-IT model card on Hugging Face https://huggingface.co/INSAIT-Institute/BgGPT-Gemma-3-27B-IT; Google DeepMind, "INSAIT creates leading Bulgarian-first LLM with Gemma 2" https://deepmind.google/models/gemma/gemmaverse/insait/.

Referenced in: Framework §2, AI-SAF-N Models pillar, AI-SAF-N Case Study (Part IV), Refining §2, §5.1

Part E. Regulatory instruments

EU AI Act — GPAI obligations applicable (2 August 2025)

Claim. Regulation (EU) 2024/1689 (the AI Act) was adopted on 13 June 2024, published in the EU's Official Journal on 12 July 2024, and came into force on 2 August 2024. The bans on certain uses and the AI-literacy duties took effect on 2 February 2025. The duties on providers of general-purpose AI (GPAI) models kicked in on 2 August 2025, and the Commission can enforce them from 2 August 2026. GPAI models already on the market before August 2025 must fully comply by 2 August 2027. Fines for GPAI providers can reach 3% of yearly worldwide revenue or €15 million, whichever is higher (AI Act Article 101).

Sources: Regulation (EU) 2024/1689 of 13 June 2024, OJ L of 12 July 2024; Future of Life Institute, "EU AI Act Implementation Timeline" https://artificialintelligenceact.eu/implementation-timeline/; Latham & Watkins, "EU AI Act: GPAI Model Obligations in Force and Final GPAI Code of Practice in Place" https://www.lw.com/en/insights/eu-ai-act-gpai-model-obligations-in-force-and-final-gpai-code-of-practice-in-place.

Referenced in: Framework §2, AI-SAF-N Regulatory pillar, AI-SAF-C Regulatory pillar, Refining §3.2, §3.8

GPAI Code of Practice (published 10 July 2025; endorsed 1 August 2025)

Claim. The General-Purpose AI Code of Practice was published by the European Commission's AI Office on 10 July 2025, and the Commission and the AI Board formally backed it on 1 August 2025. It has three chapters (Transparency; Copyright; Safety and Security) covering 12 commitments. xAI signed only the Safety and Security chapter. Amazon, Google, Microsoft, OpenAI, Anthropic and others signed all three. Meta refused to sign.

Sources: European Commission / AI Office, "The General-Purpose AI Code of Practice" https://digital-strategy.ec.europa.eu/en/policies/contents-code-gpai; Jimmy Farrell and Tekla Emborg, "An Introduction to the Code of Practice for General-Purpose AI," FLI, 14 August 2025 https://artificialintelligenceact.eu/introduction-to-code-of-practice/.

Referenced in: Framework §2, AI-SAF-N Regulatory pillar, Refining §3.2

Council of Europe Framework Convention on AI, CETS No. 225 (in force 1 November 2025)

Claim. Adopted by the Council of Europe's Committee of Ministers on 17 May 2024 and opened for countries to sign on 5 September 2024 in Vilnius. It is the first international AI treaty that is legally binding. It came into force on 1 November 2025, once the United Kingdom, France, Norway and others had ratified it (the trigger was five signatories, including at least three Council of Europe members). Early signatories include the EU, the US, the UK, Andorra, Georgia, Iceland, Norway, Moldova, San Marino and Israel. It was negotiated by the 46 Council of Europe member states, the EU, and 11 non-member states (Argentina, Australia, Canada, Costa Rica, Holy See, Israel, Japan, Mexico, Peru, US, Uruguay).

Sources: Council of Europe, CETS No. 225 (full text) https://rm.coe.int/1680afae3c; Council of Europe, "Council of Europe opens first ever global treaty on AI for signature," 5 September 2024 https://www.coe.int/en/web/portal/-/council-of-europe-opens-first-ever-global-treaty-on-ai-for-signature; BABL AI, "Council of Europe Opens Historic AI and Human Rights Treaty for Signatures," updated August 2025 https://babl.ai/council-of-europe-opens-historic-ai-and-human-rights-treaty-for-signatures/.

Referenced in: Framework §2, AI-SAF-N Regulatory pillar, Refining §3.8

Paris AI Action Summit (10—11 February 2025)

Claim. The third international AI summit, held at the Grand Palais in Paris, produced the "Statement on Inclusive and Sustainable Artificial Intelligence," signed by 61 countries and bodies including France, China, India, Japan, Australia, Canada and the EU. The United States and United Kingdom refused to sign; U.S. Vice President JD Vance criticised European AI rules in a Tuesday speech, while the UK government said the statement lacked enough substance on global governance. India, the co-chair, was named as host of the next event, the New Delhi AI Impact Summit (February 2026).

Sources: "Statement on Inclusive and Sustainable Artificial Intelligence," Paris AI Action Summit, 10—11 February 2025; Ryan Browne, "US and Britain snub international AI accord in Paris," CNBC, 11 February 2025 https://www.cnbc.com/2025/02/11/us-and-britain-snub-international-ai-accord-in-paris.html; Al Jazeera, "Paris AI summit: Why won't US, UK sign global artificial intelligence pact?," 12 February 2025 https://www.aljazeera.com/news/2025/2/12/paris-ai-summit-why-wont-us-uk-sign-global-artificial-intelligence-pact.

Referenced in: Framework §2, AI-SAF-N Regulatory pillar, Refining §3.8

Part F. Copyright and legal rulings

Bartz v. Anthropic — Summary Judgment (23 June 2025)

Claim. In Bartz v. Anthropic PBC, No. 3:24-cv-05417 (N.D. Cal.), Judge William Alsup ruled — without a full trial — that training AI models on copyrighted books was "exceedingly transformative" and counted as fair use. But he found that Anthropic downloading and keeping pirated copies from illegal book sites (Library Genesis, Pirate Library Mirror) was NOT fair use, and that part went on toward trial. On 26 August 2025, Anthropic filed notice of a proposed settlement covering the whole group of plaintiffs.

Sources: Skadden, Arps, Slate, Meagher & Flom, "Fair Use and AI Training: Two Recent Decisions," July 2025 https://www.skadden.com/insights/publications/2025/07/fair-use-and-ai-training; White & Case, "Two California district judges rule that using books to train AI is fair use," 9 July 2025 https://www.whitecase.com/insight-alert/two-california-district-judges-rule-using-books-train-ai-fair-use.

Referenced in: Framework §2, AI-SAF-N Data pillar, AI-SAF-C Data pillar, Refining §3.2

Bartz v. Anthropic — Final settlement approval (20 July 2026) [Added v0.7.1]

Claim. On 20 July 2026 Judge Araceli Martínez-Olguín (N.D. Cal.) granted final approval of the $1.5 billion class settlement — the largest copyright class settlement on record — paying roughly $3,000 per work across an estimated 500,000 works. The settlement does not release future output-based claims.

Sources: The Authors Guild, "Court Grants Final Approval of $1.5 Billion Anthropic Copyright Settlement" https://authorsguild.org/news/court-grants-final-approval-anthropic-copyright-settlement/; TechCrunch, 20 July 2026 https://techcrunch.com/2026/07/20/anthropics-landmark-1-5b-copyright-settlement-is-approved/.

Referenced in: Framework §3.1 (Data row); AI-SAF-C Data pillar; State of Play §6.

Kadrey v. Meta — Summary Judgment (25 June 2025)

Claim. In Kadrey v. Meta Platforms, Inc., No. 23-CV-03417-VC (N.D. Cal.), 2025 WL 1752484, Judge Vince Chhabria ruled in Meta's favour on fair use for the books of 13 plaintiffs, without a full trial. The ruling was narrow and tied to the specific facts: the plaintiffs had not built a strong enough case that the training hurt the market for their books. Judge Chhabria pointedly said his ruling did not give AI training on copyrighted works a green light more generally.

Sources: White & Case, "Two California district judges rule that using books to train AI is fair use," 9 July 2025 https://www.whitecase.com/insight-alert/two-california-district-judges-rule-using-books-train-ai-fair-use; Jackson Walker LLP, "Federal Courts Find Fair Use in AI Training," 11 July 2025 https://www.jw.com/news/insights-kadrey-meta-bartz-anthropic-ai-copyright/.

Referenced in: Framework §2, AI-SAF-C Data pillar, Refining §3.2

Getty Images v. Stability AI (UK) — Judgment [2025] EWHC 2863 (Ch), 4 November 2025

Claim. Mrs Justice Joanna Smith DBE of the High Court of England and Wales (Chancery Division) gave judgment in Getty Images (US) Inc & Ors v Stability AI Ltd on 4 November 2025. The court threw out Getty's secondary copyright claim — ruling that a Stable Diffusion model's internal settings (its "weights") are not an "infringing copy" under sections 22—23 of the UK's 1988 copyright law (the CDPA) — and found only limited trademark infringement involving reproduced watermarks. Getty had already dropped its main claim, that the training itself infringed copyright, because the training had happened outside the UK. This is the first significant UK ruling on copyright and AI training.

Sources: Getty Images (US) Inc & Ors v Stability AI Limited [2025] EWHC 2863 (Ch), full judgment https://www.judiciary.uk/wp-content/uploads/2025/11/Getty-Images-v-Stability-AI.pdf; Ropes & Gray, "Getty Image Loses Copyright Infringement Claim Against Stability AI," 14 January 2026
https://www.ropesgray.com/en/insights/viewpoints/102lvxe/getty-image-loses-copyright-infringement-claim-against-stability-ai-in-uks-first.

Referenced in: Framework §2, AI-SAF-N Data pillar, Refining §3.2

Part G. Research and technical references

Shumailov et al. — Model collapse (Nature, 24 July 2024)

Claim. Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson and Yarin Gal, "AI models collapse when trained on recursively generated data," Nature 631, no. 8022 (24 July 2024): 755—759, DOI 10.1038/s41586-024-07566-y. Shows that carelessly training models on content earlier models generated causes "irreversible defects" in later versions: the rare, unusual cases gradually vanish. The effect showed up across several kinds of model (large language models, variational autoencoders, and Gaussian mixture models). The authors issued a correction in Nature 640, E6 (2025), DOI 10.1038/s41586-025-08905-3.

Sources: Nature 631 (2024): 755—759 https://www.nature.com/articles/s41586-024-07566-y.

Referenced in: Framework §2, AI-SAF-N Data pillar, Refining §3.1

Borji — Note on Shumailov et al. (arXiv, October 2024)

Claim. Ali Borji, "A Note on Shumailov et al. (2024): AI Models Collapse When Trained on Recursively Generated Data," arXiv:2410.12954 [cs.LG], October 2024. Argues that model collapse is "a statistical phenomenon and may be unavoidable" any time you repeatedly fit a pattern to data and then draw new samples from it.

Sources: arXiv:2410.12954 https://arxiv.org/abs/2410.12954.

Referenced in: Framework §2, AI-SAF-N Data pillar, Refining §3.1

DeepSeek-V3 Technical Report (27 December 2024)

Claim. DeepSeek-AI team, "DeepSeek-V3 Technical Report," arXiv:2412.19437 (initial 27 December 2024, updated 18 February 2025). DeepSeek-V3 is a "Mixture-of-Experts" model — one that has 671 billion parameters in total but only switches on 37 billion at a time — trained on 14.8 trillion tokens of text. The reported training cost: 2.664 million H800 GPU-hours for the main training, 119,000 for extending its context length, and 5,000 for final tuning, for a total of 2.788 million GPU-hours; at $2 per H800 GPU-hour, that comes to about $5.576 million. That figure leaves out earlier research, trial-and-error experiments, and the cost of the underlying hardware spread over time. Its technical innovations include FP8 mixed-precision training, a way of balancing the load without an extra penalty term, and predicting several tokens at once.

Sources: arXiv:2412.19437 https://arxiv.org/abs/2412.19437.

Referenced in: Framework §2, AI-SAF-N Models pillar, Refining §3.2

Stanford AI Index 2026 (April 2026)

Claim. The 2026 edition of the Stanford Institute for Human-Centered AI (HAI) AI Index Report was published in April 2026. The ninth yearly edition; it covers research, technical performance, responsible AI, economics, science, medicine, education, policy, and public opinion. The key 2026 findings the framework draws on: the capability gap between the top US and Chinese models has all but closed; 44 countries now run government-backed supercomputers; 80 of the 95 most notable 2025 models were released without their training code; the Foundation Model Transparency Index average score dropped to 40 (from 58); and almost the entire global AI industry still depends on TSMC's chip factories in Taiwan.

Sources: Stanford HAI, 2026 AI Index Report https://hai.stanford.edu/ai-index/2026-ai-index-report.

Referenced in: Framework §3, AI-SAF-N all pillars, AI-SAF-C all pillars, Refining §1.3, §3.7

Part H. Benchmark frameworks (peer indices)

These are the rival frameworks named in the AI-SAF comparison tables. Each is its own tool with its own scope and method; AI-SAF sees itself as filling a different gap from most of them (measuring whether a setup can be reversed) rather than competing head-on.

IMF AI Preparedness Index (AIPI)

Claim. Covers 174 economies across four equally-weighted areas: digital infrastructure, people and labor-market policy, innovation and economic links, and regulation and ethics. The underlying measures are each rescaled to a 0—1 range (using the min-max method), then the four areas are averaged. Wealthy economies average 0.68, emerging markets

0.46, and low-income countries 0.32. The data is pulled together from the Fraser Institute, ILO, ITU, UN, UNCTAD, Universal Postal Union, World Bank, and World Economic Forum.

Sources: IMF Datamapper: AI Preparedness Index (AIPI) https://www.imf.org/external/datamapper/datasets/AIPI; AIPI Methodology Note https://www.imf.org/external/datamapper/AIPINote.pdf; Kristalina Georgieva et al., "Mapping the World's Readiness for Artificial Intelligence," IMF Blog, 25 June 2024 https://www.imf.org/en/Blogs/Articles/2024/06/25/mapping-the-worlds-readiness-for-artificial-intelligence-shows-prospects-diverge.

Referenced in: Framework §3 comparison table, Refining §1.3

Tortoise Global AI Index (5th edition, 19 September 2024)

Claim. Covers 83 countries using 122 measures grouped under 3 headings (Implementation, Innovation, Investment) and 7 sub-headings (talent, infrastructure, operating environment, research, development, commercial ecosystem, government strategy). It draws on 24 public and private data sources. The United States ranked 1st, China 2nd, Singapore 3rd, the United Kingdom 4th, and France 5th. Saudi Arabia came top on the government-strategy sub-heading.

Sources: Tortoise Media, Global AI Index 2024 https://www.tortoisemedia.com/data/global-ai; Methodology Report (September 2024) https://www.tortoisemedia.com/_app/immutable/assets/AI-Methodology-2409.BGTLUPC-.pdf.

Referenced in: Framework §3 comparison table, AI-SAF-N comparison, Refining §1.3

MacroPolo Global AI Talent Tracker 2.0 (March 2024, NeurIPS 2022 data)

Claim. Published by the Paulson Institute. Based on a sample of 186 papers and 867 authors from the NeurIPS 2022 conference (95% confidence, give or take 7%). Key findings: 75% of the top AI researchers working at US institutions did their undergraduate degrees in the US or China (up from 58% in 2019); China produces 47% of the world's top AI researchers, counted by where they did their undergraduate degree (up from 29% in 2019); the US is still the top destination for elite talent; and about 42% of top researchers work abroad (down from 55% in 2019), meaning fewer are moving around overall. Of Chinese AI undergraduates, 51.35% go on to graduate study at home, and 30.96% stay in China for that work. In India, one-fifth of AI researchers now stay in India after graduating (up from almost none in 2019).

Sources: MacroPolo, Global AI Talent Tracker 2.0 https://archivemacropolo.org/interactive/digital-projects/the-global-ai-talent-tracker/; Methodology for Global AI Talent Tracker 2.0 https://archivemacropolo.org/interactive/digital-projects/the-global-ai-talent-tracker/methodology-for-global-ai-talent-tracker-2/; Zeyi Yang, "Four things you need to know about China's AI talent pool," MIT Technology Review, 27 March 2024 https://www.technologyreview.com/2024/03/27/1090182/ai-talent-global-china-us/.

Note. An earlier draft of the refinement white paper called this “Global AI Talent Tracker 3.0” — the correct name is 2.0, as shown above.

Referenced in: Framework §3, AI-SAF-N Human Capital pillar, AI-SAF-C Human Capital pillar, Refining §3.7

NIST AI Risk Management Framework (AI RMF 1.0) and GenAI Profile

Claim. National Institute of Standards and Technology, Artificial Intelligence Risk Management Framework, NIST AI 100-1 (Gaithersburg, MD: NIST, January 2023). A voluntary US framework built around four jobs: GOVERN, MAP, MEASURE, MANAGE. The Generative AI Profile (NIST AI 600-1, July 2024) tailors it for generative-AI systems, with 12 categories of risk. In practice, it is the go-to reference for managing AI risk in the US private sector.

Sources: NIST AI RMF 1.0 https://www.nist.gov/itl/ai-risk-management-framework; AI RMF Playbook https://airc.nist.gov/AI_RMF_Knowledge_Base/Playbook.

Referenced in: Framework §3, AI-SAF-C Culture & Governance pillar (primary basis), Refining §4.1

ISO/IEC 42001:2023 — AI Management System

Claim. Published by ISO and IEC in December 2023. The first international AI management standard a company can actually be certified against. It is laid out the same way as ISO 27001 and ISO 9001 (the common “Annex SL” structure): Clauses 4—10 (Context, Leadership, Planning, Support, Operation, Performance Evaluation, Improvement) plus Annex A controls covering reducing bias, transparency, accountability, and data governance. Certification usually lasts three years, with yearly check-up audits in between. Microsoft 365 Copilot was certified in 2025.

Sources: ISO/IEC 42001:2023 standard page https://www.iso.org/standard/42001; ANAB, “ISO/IEC 42001: Artificial Intelligence Management Systems,” ANAB Blog https://blog.ansi.org/anab/iso-iec-42001-ai-management-systems/; Deloitte, “ISO 42001 Standard for AI Governance and Risk Management” https://www.deloitte.com/us/en/services/consulting/articles/iso-42001-standard-ai-governance-risk-management.html.

Referenced in: Framework §3, AI-SAF-C Culture & Governance pillar, AI-SAF-C Part V crosswalk, Refining §4.2

GAIA-X Labelling Framework

Claim. GAIA-X European Association for Data and Cloud AISBL, Brussels. Its Labelling Framework version 2.0 and later updates set out six groups of criteria (data protection, cybersecurity, transparency, portability and interoperability, flexibility, and operational requirements) across three levels (L1, L2, L3). Level L3 — effectively fully sovereign — requires the same protection from foreign-reach laws as SecNumCloud. It lines up directly with the Infrastructure scoring bands in AI-SAF-N and AI-SAF-C.

Sources: GAIA-X Association, "Labelling Framework" documents https://gaia-x.eu/.

Referenced in: Framework §3, AI-SAF-C Infrastructure pillar, AI-SAF-C Part V crosswalk, Refining §4.3

OECD.AI Policy Observatory

Claim. Tracks national AI strategies, spending on computing power, and regulatory moves across OECD members and partner countries. It focuses on governance rather than on raw capability. It organises its data around the five OECD AI Principles (inclusive growth, human rights, transparency, robustness, and accountability). A useful companion to the indices that focus on capability.

Sources: OECD.AI https://oecd.ai/.

Referenced in: Framework §3 comparison table, Refining §1.3

CSET / ETO (Emerging Technology Observatory, Georgetown University)

Claim. The Center for Security and Emerging Technology's Emerging Technology Observatory runs several detailed dashboards: the Country Activity Tracker (CAT), the Private AI-Related Activity Tracker (PARAT), the Supply Chain Explorer, and AGORA. Together they are the most detailed public window into the chip and AI-research supply chains. Recommended as a primary source for scoring the AI-SAF Hardware and Manufacturing building blocks.

Sources: ETO dashboards https://eto.tech/.

Referenced in: Framework §3 comparison table, AI-SAF-N Hardware pillar, AI-SAF-N Manufacturing pillar, Refining §1.3

Other peer frameworks referenced

Claim. Other frameworks worth consulting: the Stanford HAI AI Index (see Part G); the Oxford Insights Government AI Readiness Index (188 countries, focused on government capacity); the ECFR Technology Sovereignty Tracker (which maps the dependencies of the 27 EU member states); Bertelsmann Stiftung's Sustainable Governance Indicators and EuroStack work; the Nextcloud Digital Sovereignty Index (EU-focused, centred on open-source hosting); the UNCTAD Technology and Innovation Report 2025; MIT CISR's enterprise AI research; and the BCG, McKinsey, and Gartner enterprise AI-maturity models.

Sources: Oxford Insights Government AI Readiness Index https://oxfordinsights.com/ai-readiness/; UNCTAD Technology and Innovation Report https://unctad.org/publication/technology-and-innovation-report-2025.

Referenced in: Framework §3 comparison table, Refining §1.3

Part J. Claims not yet individually verified

This section lists the kinds of claims in the framework documents that we did not check one by one against original sources this time around. They are not in doubt — they simply have not yet been traced back to authoritative sources in this pass. A future update of this Evidence Base should cover them. 1. Specific multilingual LLM parameter counts and dataset sizes Beyond BgGPT (verified in Part D), Mistral (verified in Part D), DeepSeek-V3 (verified in Part G), and general Llama references: Llama 4 Scout and Maverick specifications; Gemma 3 licence details beyond those in BgGPT model cards; Qwen 3.5 and 3.6; Falcon 3; Cohere Aya-Expanse; EuroLLM-9B and EuroLLM-22B; Teuken-7B; EXAONE 4.5; SEA-LION v3; Sarvam-M; TAIDE; Fugaku-LLM; Meltemi; Poro-34B. 2. Chip architecture details beyond Rubin, H20/H200, Intel Magdeburg Google TPU Ironwood; AWS Trainium3; Microsoft Maia 200; Meta MTIA RISC-V; Cerebras generations; Groq acquisition / integration with NVIDIA (Groq 3 LPX); Tenstorrent funding; SambaNova acquisition; Rebellions funding; SiPearl Rhea1 and Rhea2; EU DARE RISC-V project; Huawei Ascend 910C and 950PR specifications. 3. TSMC, ASML, CoWoS and HBM4 supply figures TSMC N2 wafer pricing (reported ranges $25,000–$27,000 vary across sources); CoWoS advanced packaging capacity figures; HBM4 allocations by supplier (SK Hynix, Samsung, Micron); ASML EUV and High-NA EUV unit shipments for 2025 and 2026. 4. CPU roadmap details AWS Graviton5; Arm AGI CPU; AMD EPYC Venice (Zen 6); Intel Clearwater Forest. (NVIDIA Vera CPU is verified in Part I.) 5. Regulatory instruments beyond those in Part E UN Global Digital Compact (September 2024); UK AI Safety Institute rebrand to AI Security Institute (February 2025); US AI Safety Institute rebrand to CAISI (June 2025). (The New Delhi AI Impact Summit outcomes, listed here in earlier revisions, were verified in v0.7.1 and moved to Part E.) 6. Dataset and benchmark details INCLUDE benchmark; Global-MMLU; CulturaX (~6.3T tokens claimed); HPLT v2; FineWeb-2; Meltemi 40B Greek tokens claim; Poro-34B 1T tokens claim. 7. Specific commercial deals BNP Paribas / Mistral AI partnership (February 2024); Cerebras / OpenAI reported $10B partnership (January 2026); various other announced partnerships cited as context. If you are making decisions based on any of these specific figures, check them against each project's own documentation first.

Version History

Version 0.7.1 — August 2026

A revision in three layers, responding to the August 2026 evidence review.

Evidence and consistency (what a 0.5.8 would have carried). Evidence Base updated to August 2026 with eleven added or extended entries: the Fable 5 / Mythos 5 restoration (30 June – 1 July), US chip-export posture codification (January–July 2026), DeepSeek-V4 with day-0 Ascend adaptation, the Nexperia standoff (a pre-snapshot gap), the June–July rare-earth enforcement wave, the AI Gigafactories call, the first EU sovereignty-criteria cloud procurement, the Digital Omnibus deferral with on-schedule GPAI enforcement, the CADA and Chips Act 2.0 proposals, the New Delhi Declaration (promoted from Part J), and the Bartz final settlement approval. The Fable/Mythos arc is reconciled everywhere it is cited.

The v0.5.7 L0–L4 relabelling is completed in the BgGPT case study and the corporate guide (INSAIT profile Models 12 → 11; totals adjusted). MacroPolo tracker corrected to 2.0 throughout; CAISI naming standardised; a stale reference in the Evidence Base verification note removed.

Methodology. Every sub-indicator now carries an R (resilience) or A (autonomy) tag, and assessments report the two subtotals alongside the headline (Framework §1.1) — the two-axes definition now reaches the arithmetic. AI-SAF-C Models: portfolio diversification is rescored as portfolio-and-jurisdiction diversification (a single-jurisdiction portfolio caps at 2 of 5), reflecting the June 2026 lesson that vendor count is not the binding risk. Version Deadlock taxonomy gains the restoration-is-not-recovery rule (§3.5). The Infrastructure anchors add the proposed CADA Cloud Sovereignty Framework alongside SecNumCloud 3.2 and GAIA-X L3. §4.3 now states plainly which rigour checks are design commitments rather than completed exercises, and Part V owns the limits of additive aggregation. Modifier discipline is enforced: the national worked examples apply all modifiers in a fixed order, and the ad-hoc bonuses in earlier revisions (France "+5 rule-setting," UAE "+5 energy") are removed — France recomputes to ~70 (from ~71), the UAE to ~57 (from ~62), Bulgaria is unchanged at ~46.

Structure and figures. The package is reorganised: the corporate variant (AI-SAF-C) now leads as Document 2; dated market material moves from the field guides into a new Document 4, State of Play, refreshed each revision; the Evidence Base becomes Document 5; the three glossaries merge into one master glossary in Document 1; and repeated passages (the "same pillar, opposite meaning" argument, the CrowdStrike rationale) are stated once. The package is roughly a third shorter than v0.5.7 with no scoring content removed. Both figures are redrawn: the architecture diagram with the correct weights, fixed modifier order, and R/A tags; the trajectory chart in the package palette with the stale pre-rename axis label corrected.

Pillar weights are unchanged. Toward 1.0: publication of the Monte Carlo sensitivity runs and PCA cross-checks (§4.3), and at least two independent assessments landing within one band of the author's.

Version 0.5.7 — July 2026. Responded to the first external review: Models ladder rescaled to the 16-point weight (L0–L4 = 0/4/11/13/16); pooled EU compute credited in Local compute (≤2 of 4 points); front matter consolidated; residual inconsistencies corrected.

Version 0.5.6 — June 2026. Added the data-egress dimension to the Data pillar in both field guides; pillar totals held by redistribution.

Version 0.5.5 — June 2026. Added the Anthropic Fable 5 / Mythos 5 suspension (12 June 2026) as a flagship case; no scores or methodology changed.

Version 0.5.4 — May 2026. First public release of the document package.

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AI Strategic Autonomy Framework: Assessment Framework

Version 0.7.1 · August 2026

Factual snapshot: April 2026, with an August 2026 evidence update (entries marked [Added v0.7.1] in the Evidence Base; Document 4, State of Play, reflects August 2026 throughout). The state-of-play material ages quickly; the package is revised on a stated cadence, and each revision updates the snapshot date here and on the cover.

AI assistance disclosure: Drafting, editing, and evidence-base compilation were assisted by Claude (Anthropic). All analysis, judgments, and final content are the sole responsibility of the author.

Suggested citation: Alexandrov, G. (2026). *AI Strategic Autonomy Framework: Assessment Framework* (Version 0.7.1) [Document package]. https://doi.org/10.5281/zenodo.22178006

Future updates will be recorded in the Version History. Small fixes (typos, broken links) keep the same version number with a dated note; changes to scoring, pillar definitions, or method get a new version number.

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